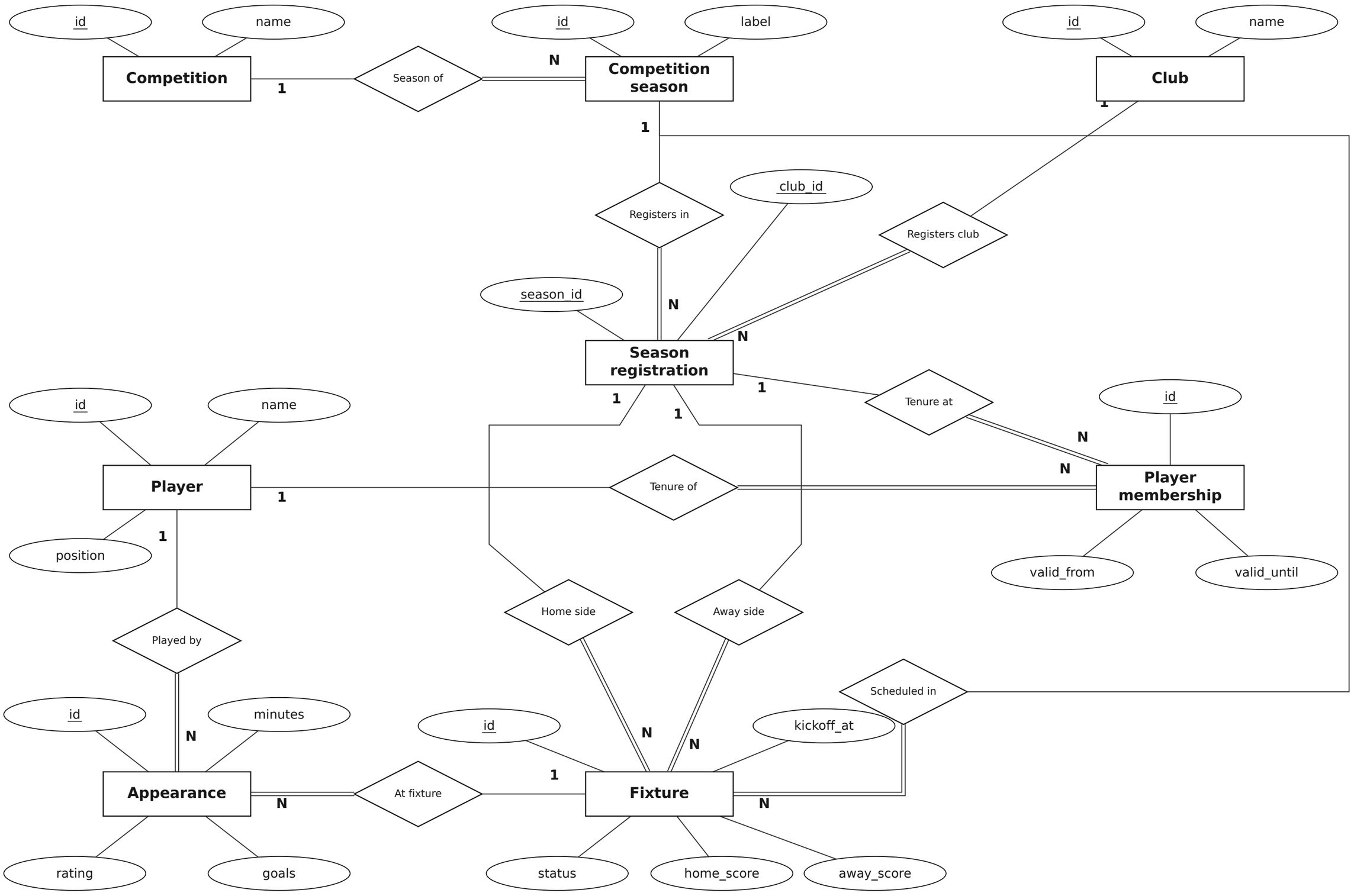
FOOTBALL ENTITY–RELATIONSHIP MODEL

Chen business overview · complete attributes and supporting entities continue in the five ER sheets



1/N are relationship maxima; double lines show total participation. Both underlined registration fields form one combined key.

This overview selects business attributes. The complete ER views include nullable context, provenance, snapshots, forecasts and services.

The full relational derivation preserves every stored attribute, alternate key and ordered FK group; goal and rating observations remain nullable.

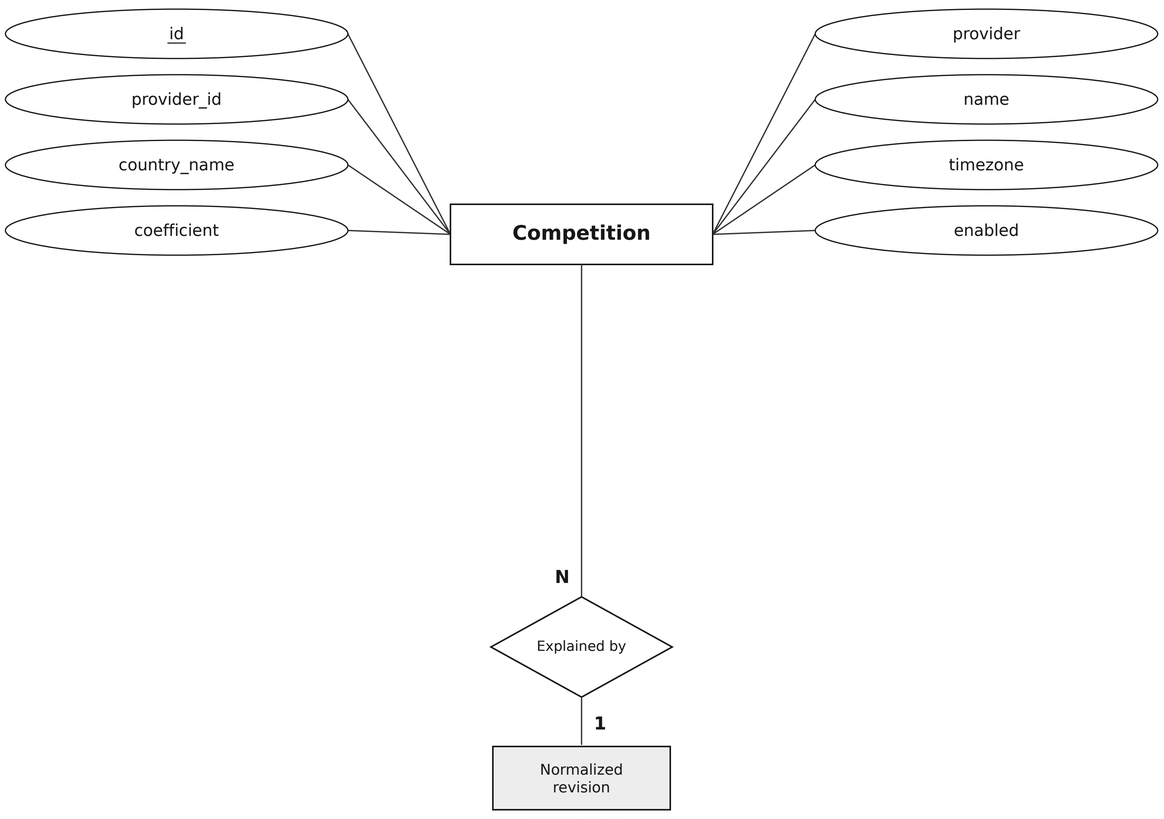
CHEN ER MODEL · 1/5 · Football and registration

Rectangles: entities · diamonds: relationships · ovals: attributes · solid underline: chosen key

1/N give maximum ratios; double lines give mandatory participation. Gray boxes repeat an existing entity.

sports.competitions

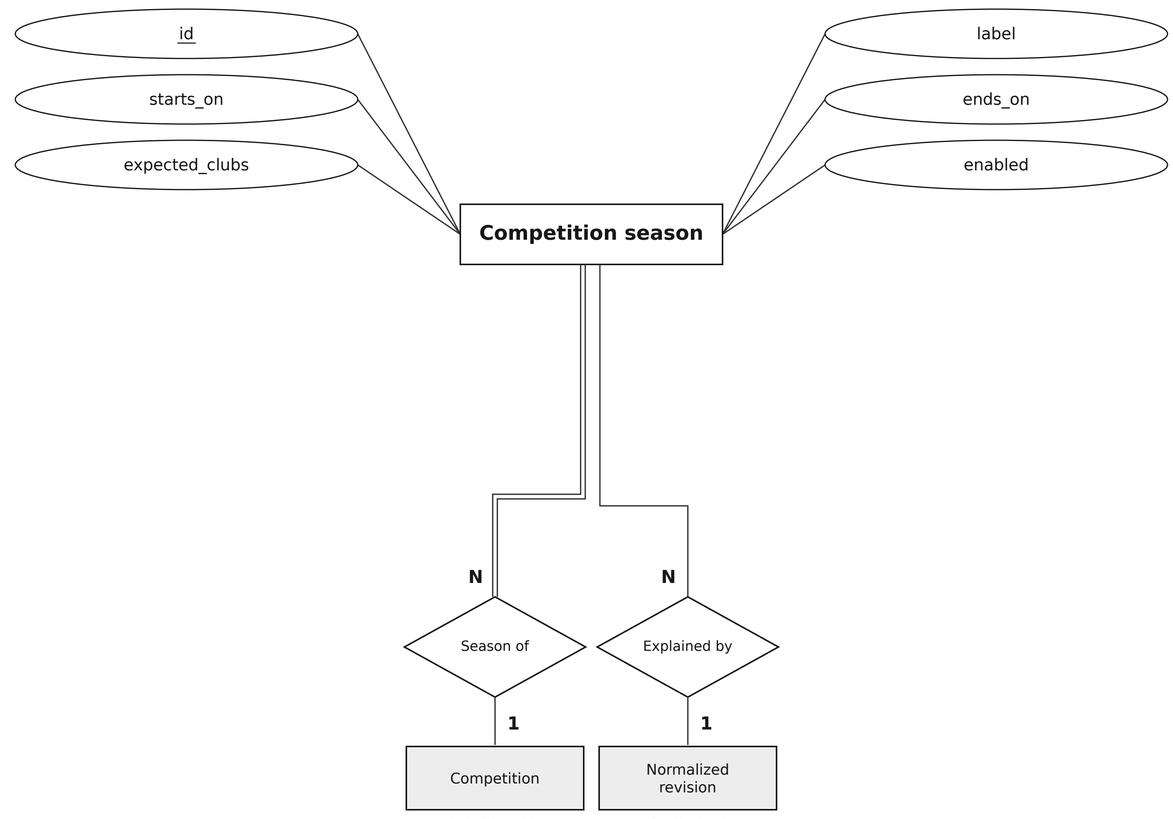
Entity and its stored scalar attributes



REFERENCE · ER 3

sports.competition_seasons

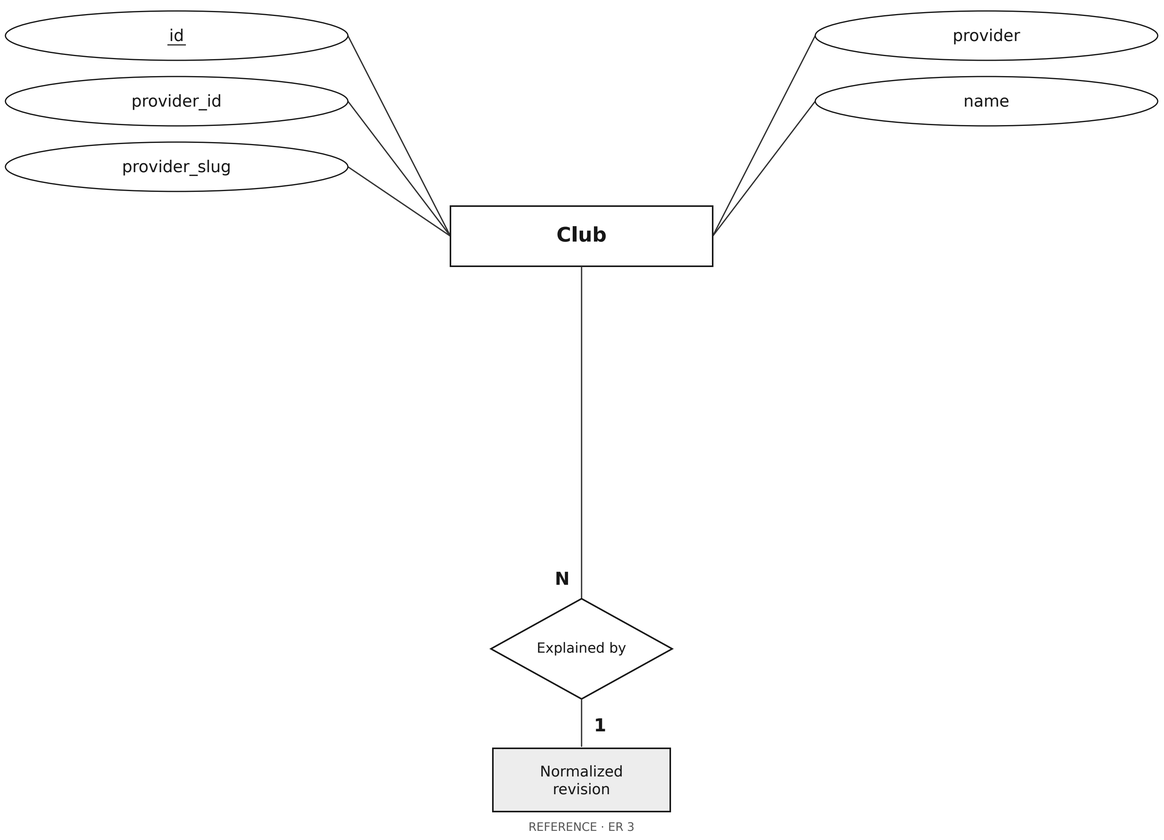
Entity and its stored scalar attributes



REFERENCE · ER 1 REFERENCE · ER 3

sports.clubs

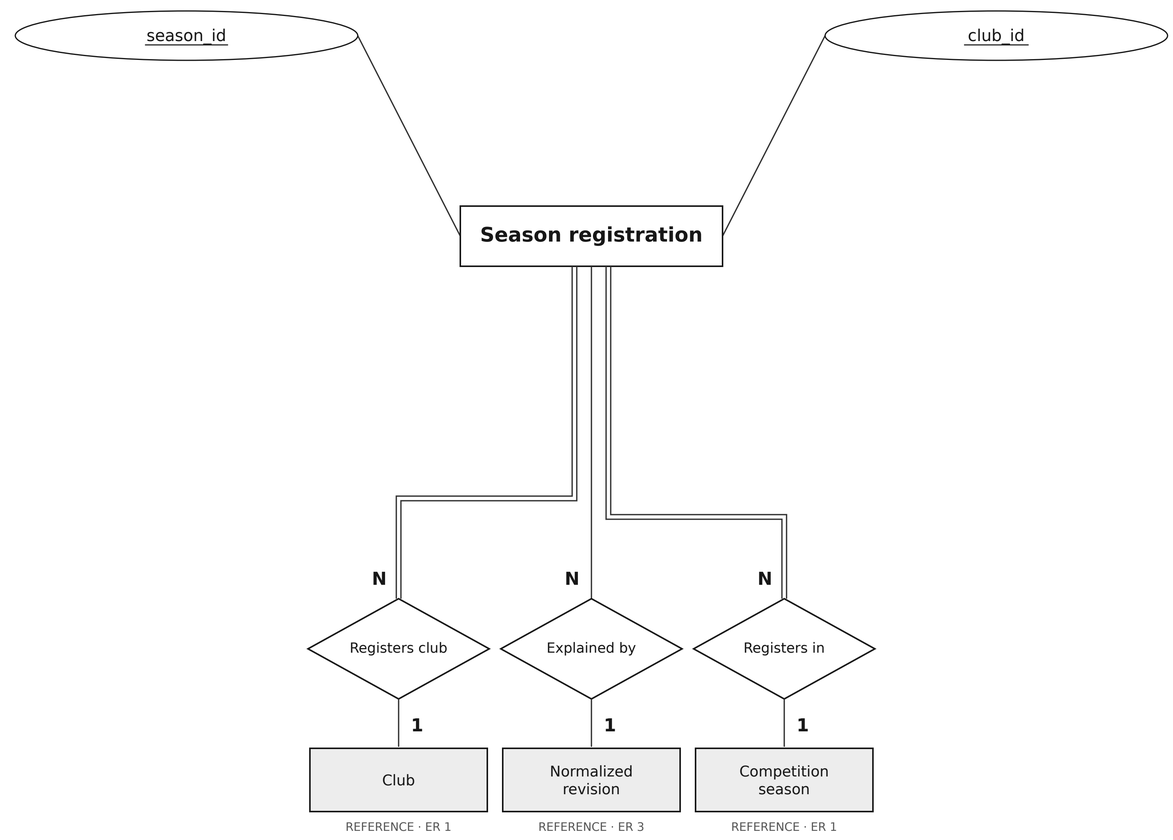
Entity and its stored scalar attributes



REFERENCE · ER 3

sports.season_clubs

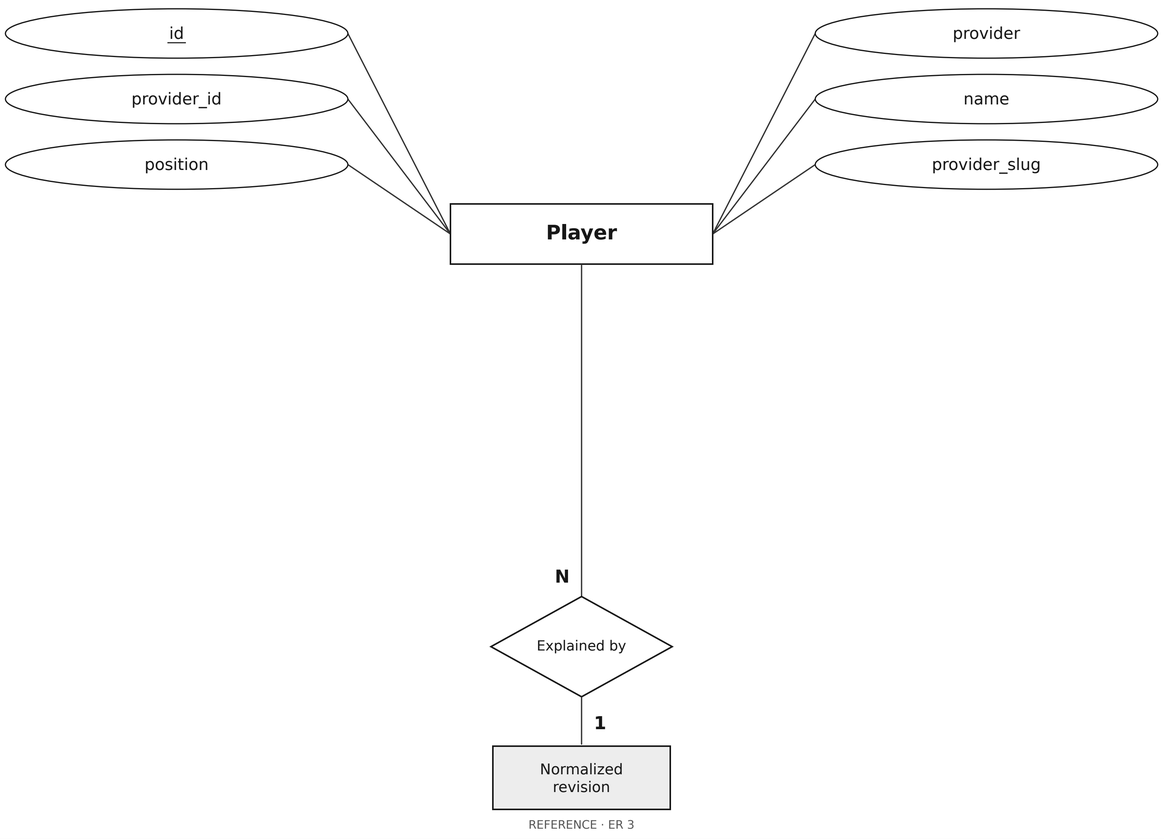
Composite-key association



REFERENCE · ER 1 REFERENCE · ER 3 REFERENCE · ER 1

sports.players

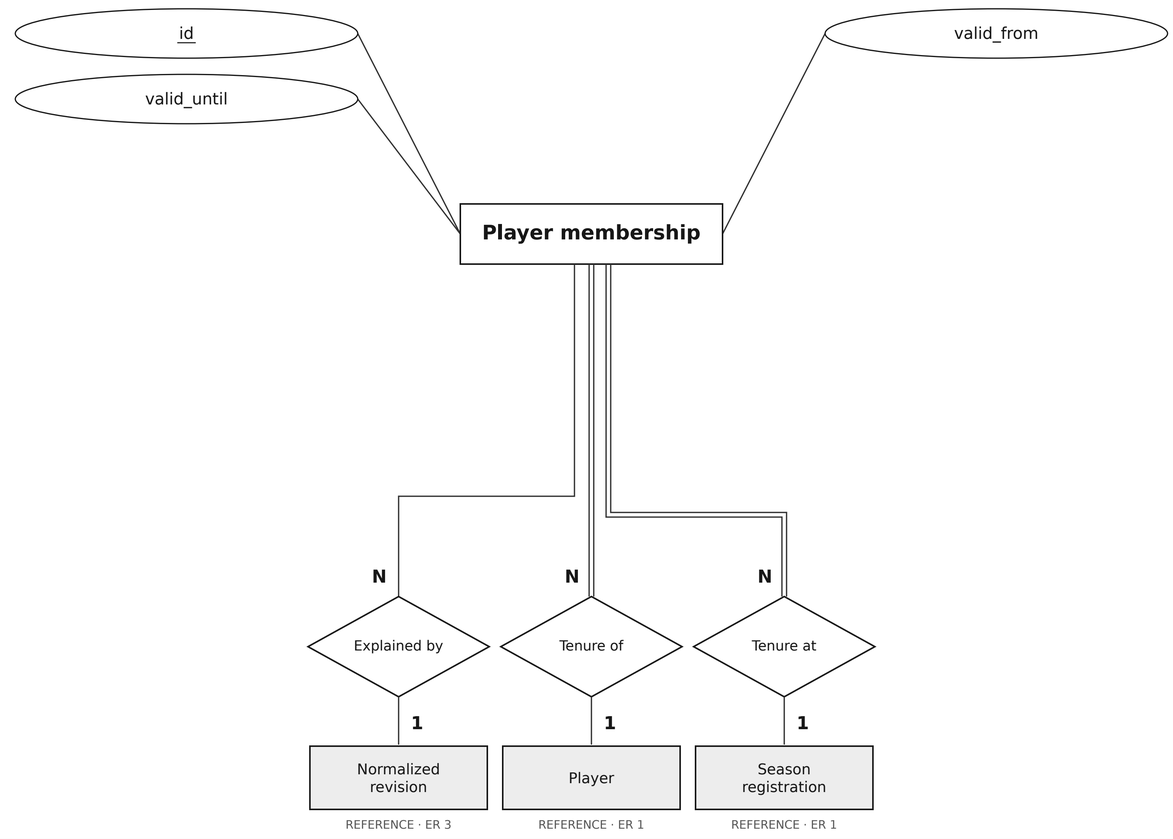
Entity and its stored scalar attributes



REFERENCE · ER 3

sports.player_memberships

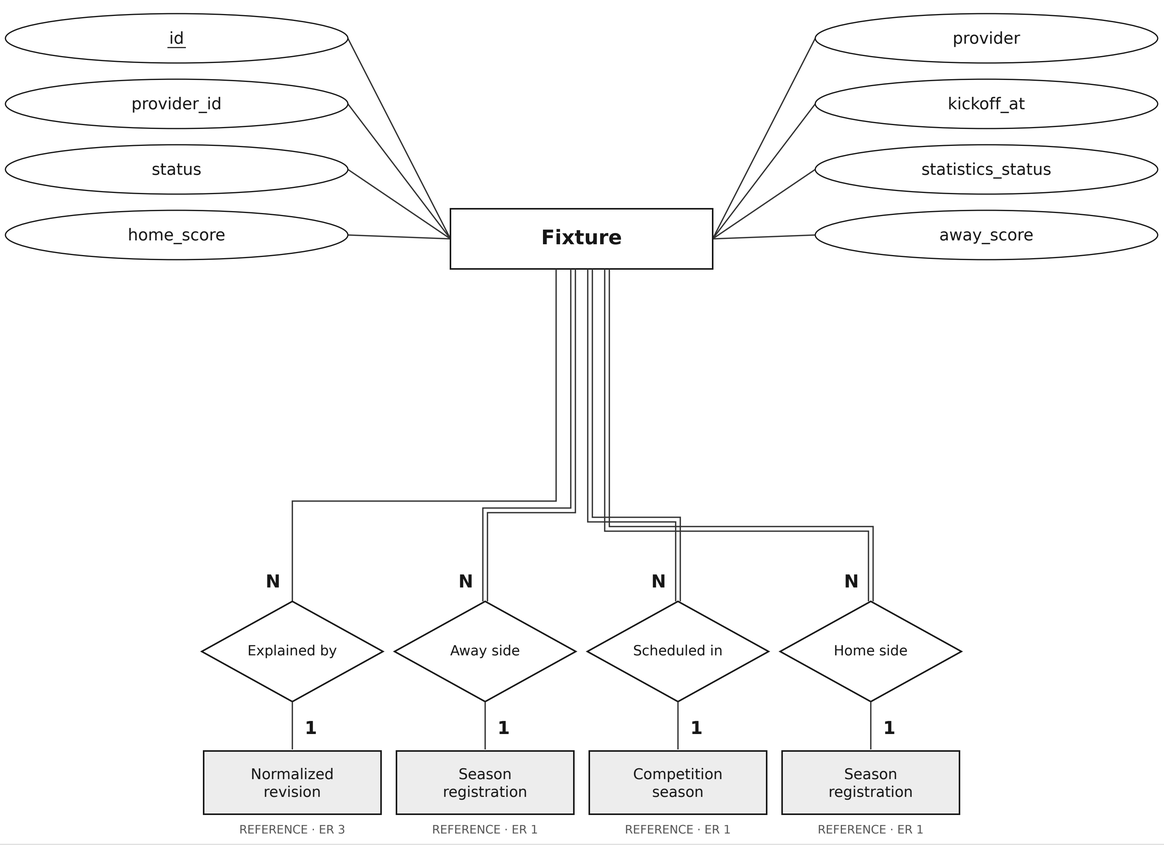
Entity and its stored scalar attributes



REFERENCE · ER 3 REFERENCE · ER 1 REFERENCE · ER 1

sports.fixtures

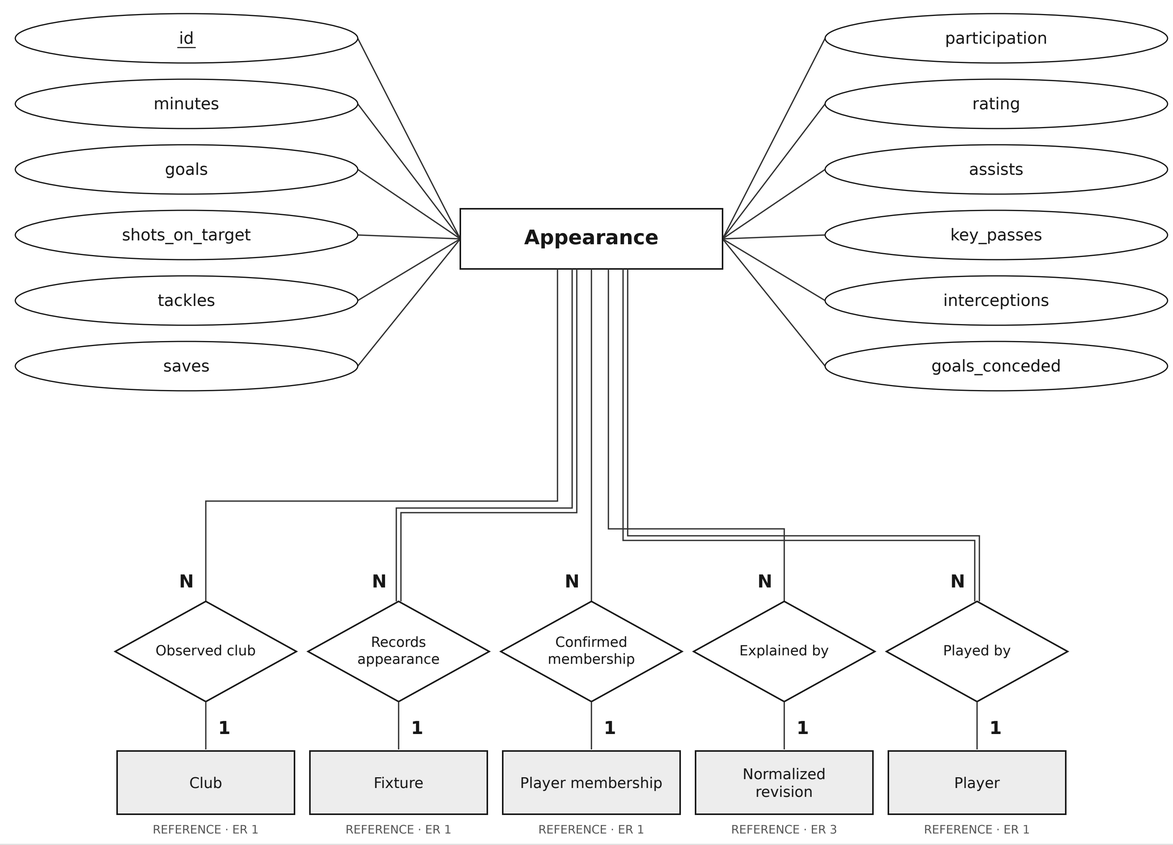
Entity and its stored scalar attributes



REFERENCE · ER 3 REFERENCE · ER 1 REFERENCE · ER 1 REFERENCE · ER 1

sports.appearances

Entity and its stored scalar attributes



REFERENCE · ER 1 REFERENCE · ER 1 REFERENCE · ER 1 REFERENCE · ER 3 REFERENCE · ER 1

Double rectangle + double diamond: weak identification. Dashed underline: partial key. Dashed oval: derived.

club_id identifies a standings row only within its owner snapshot; goal_difference = goals_for − goals_against.

These local views join on repeated entity names. FK column groups, alternate keys and conditional rules: relational-schema.md.

Current diamonds map to shared-owner-key head relations. No multivalued attribute or extra entity is invented.

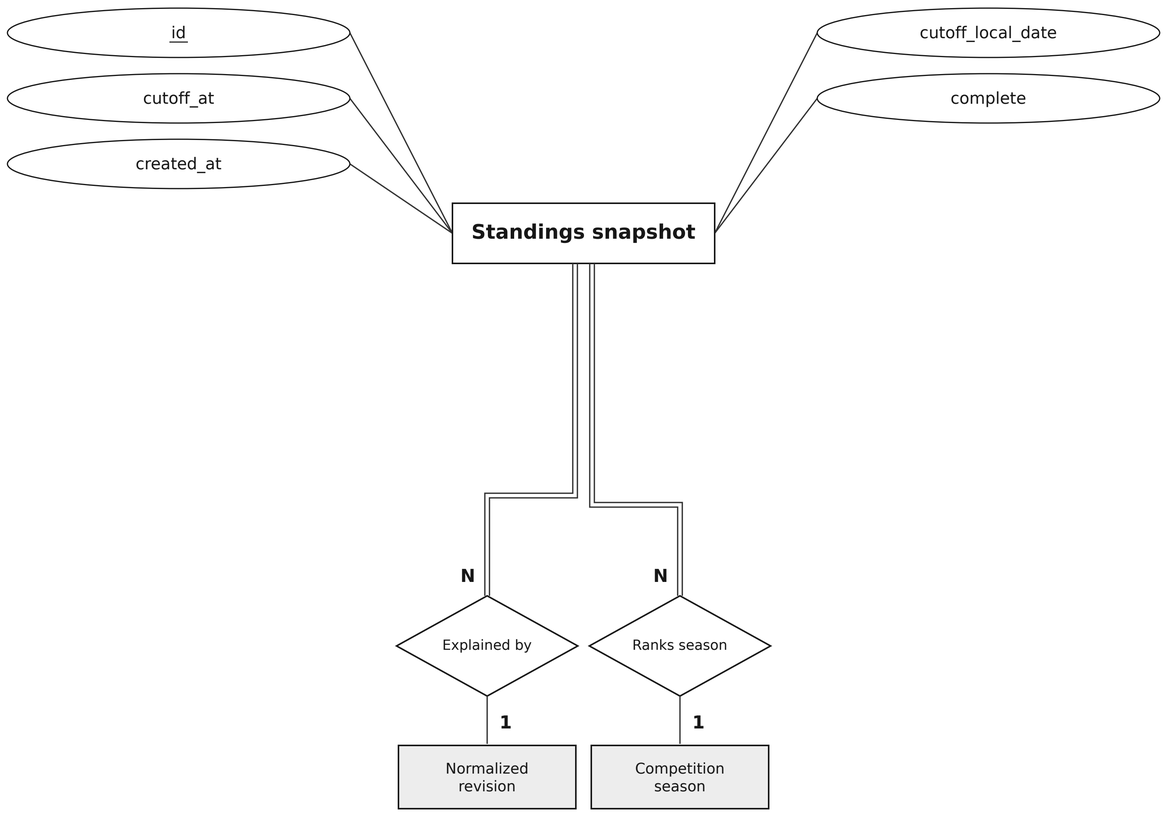
CHEN ER MODEL · 2/5 · Standings and roster observations

Rectangles: entities · diamonds: relationships · ovals: attributes · solid underline: chosen key

1/N give maximum ratios; double lines give mandatory participation. Gray boxes repeat an existing entity.

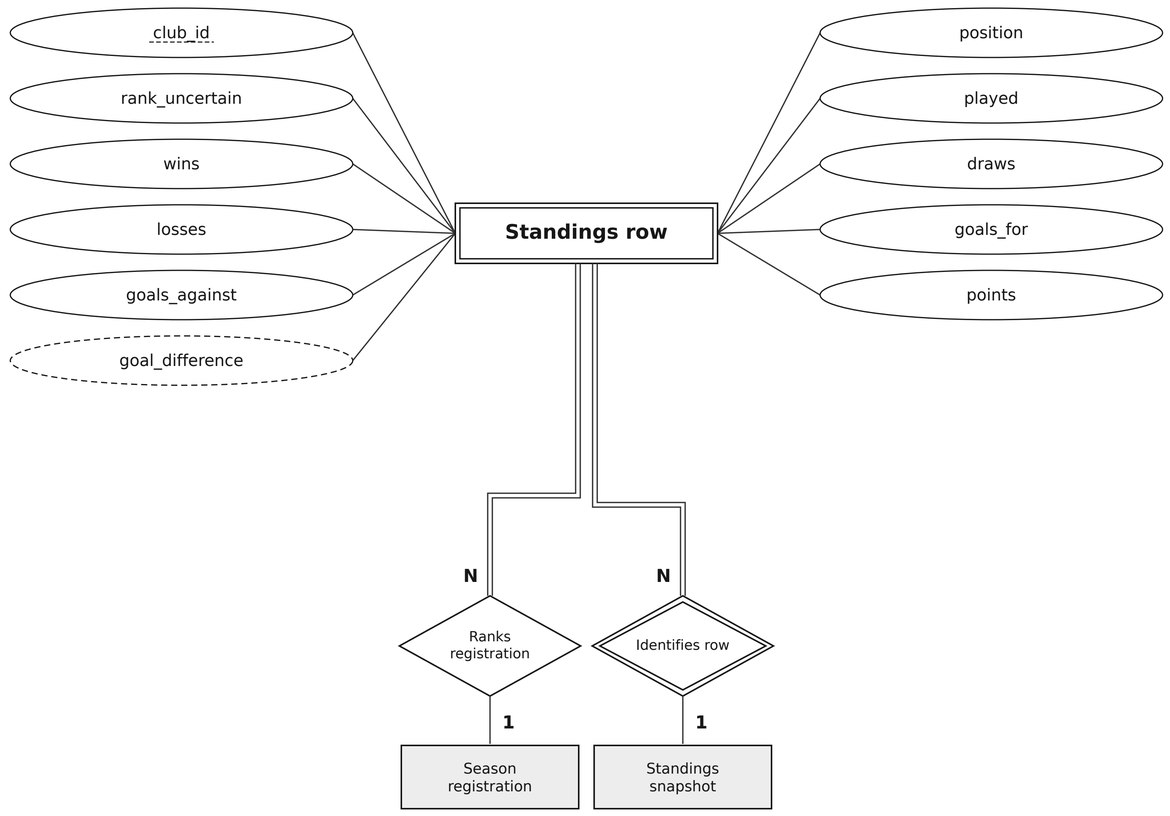
sports.standings_snapshots

Entity and its stored scalar attributes



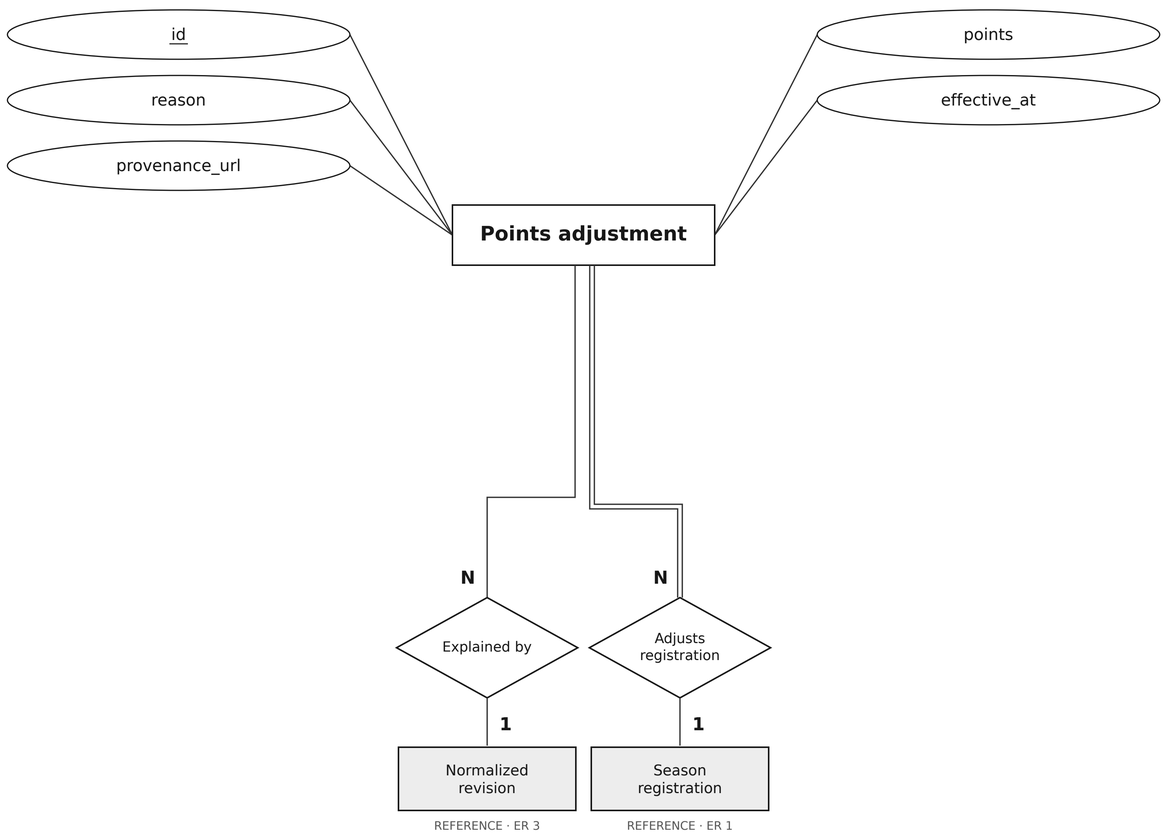
sports.standings_rows

Weak entity: owner key + club_id



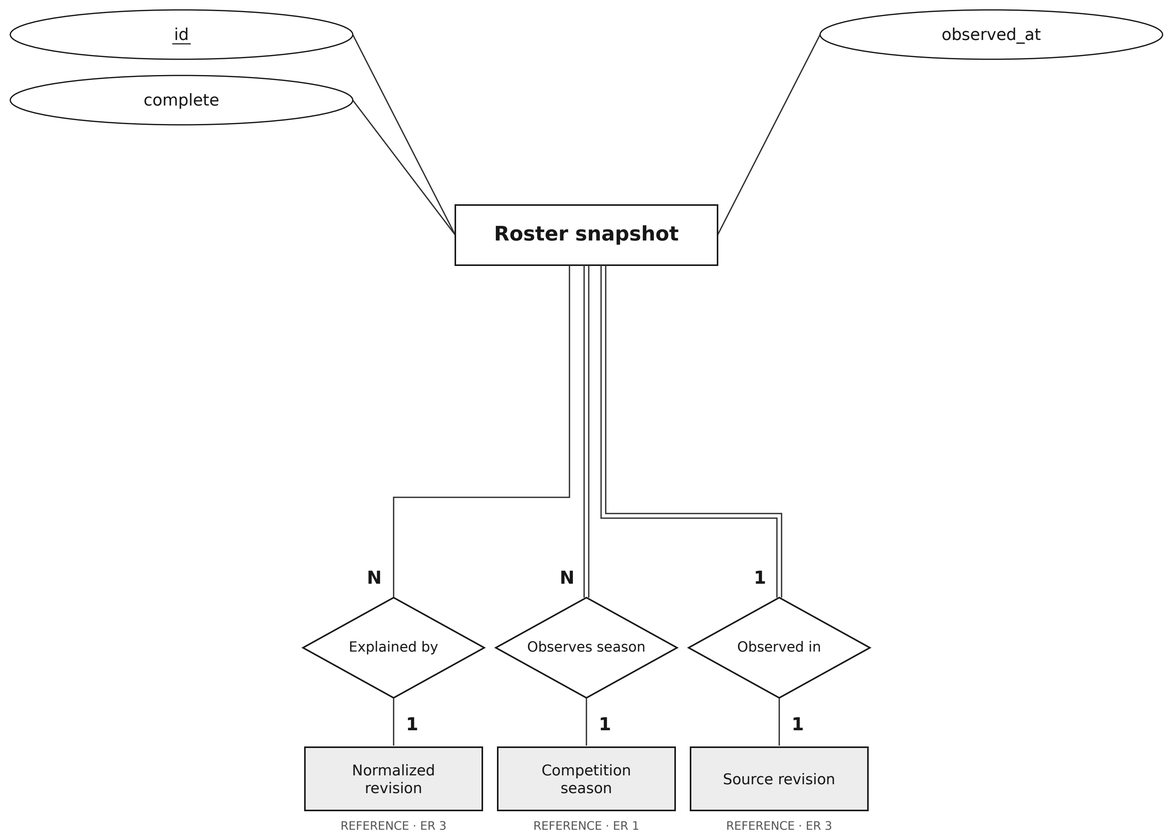
sports.points_adjustments

Entity and its stored scalar attributes



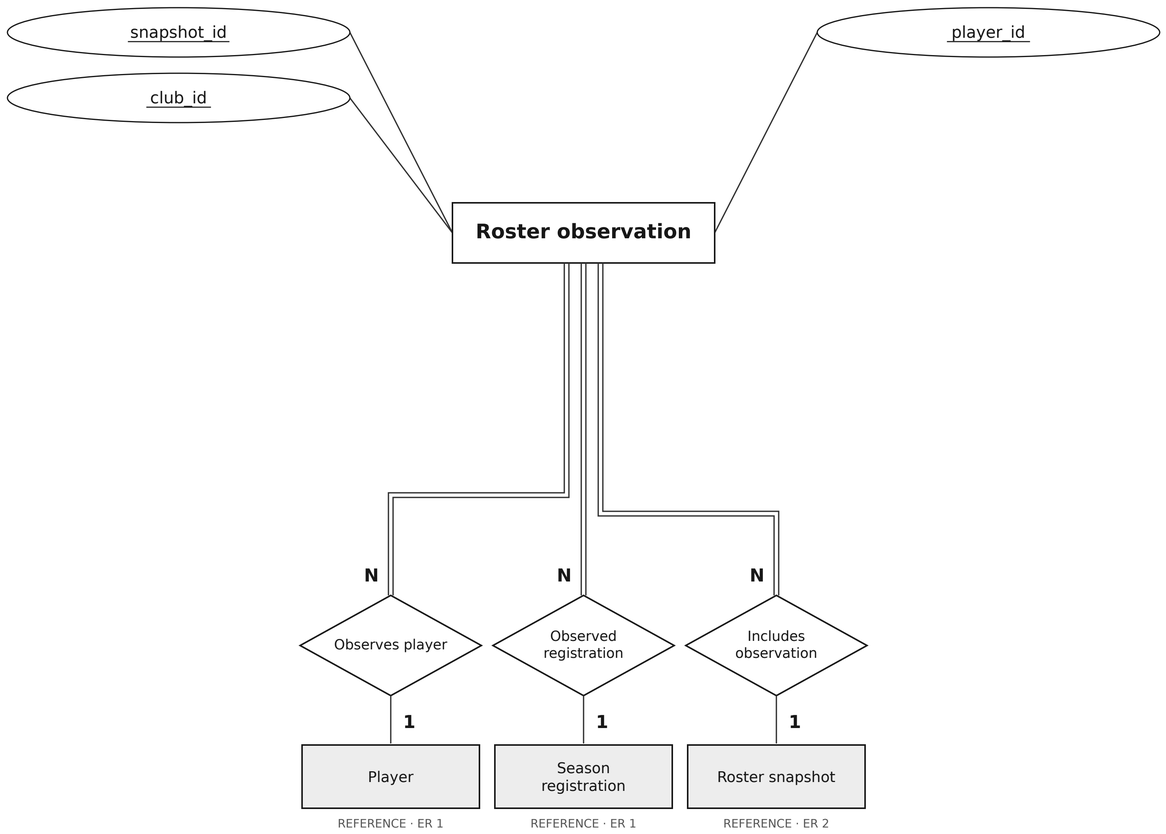
sports.roster_snapshots

Entity and its stored scalar attributes



sports.roster_entries

Composite-key association



Double rectangle + double diamond: weak identification. Dashed underline: partial key. Dashed oval: derived.

club_id identifies a standings row only within its owner snapshot; goal_difference = goals_for − goals_against.

These local views join on repeated entity names. FK column groups, alternate keys and conditional rules: relational-schema.md.

Current diamonds map to shared-owner-key head relations. No multivalued attribute or extra entity is invented.

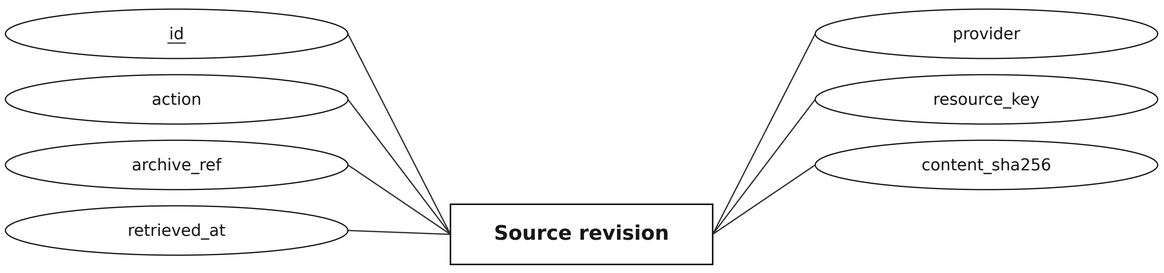
CHEN ER MODEL · 3/5 · Provenance, supplements and forecasts

Rectangles: entities · diamonds: relationships · ovals: attributes · solid underline: chosen key

1/N give maximum ratios; double lines give mandatory participation. Gray boxes repeat an existing entity.

sports.source_revisions

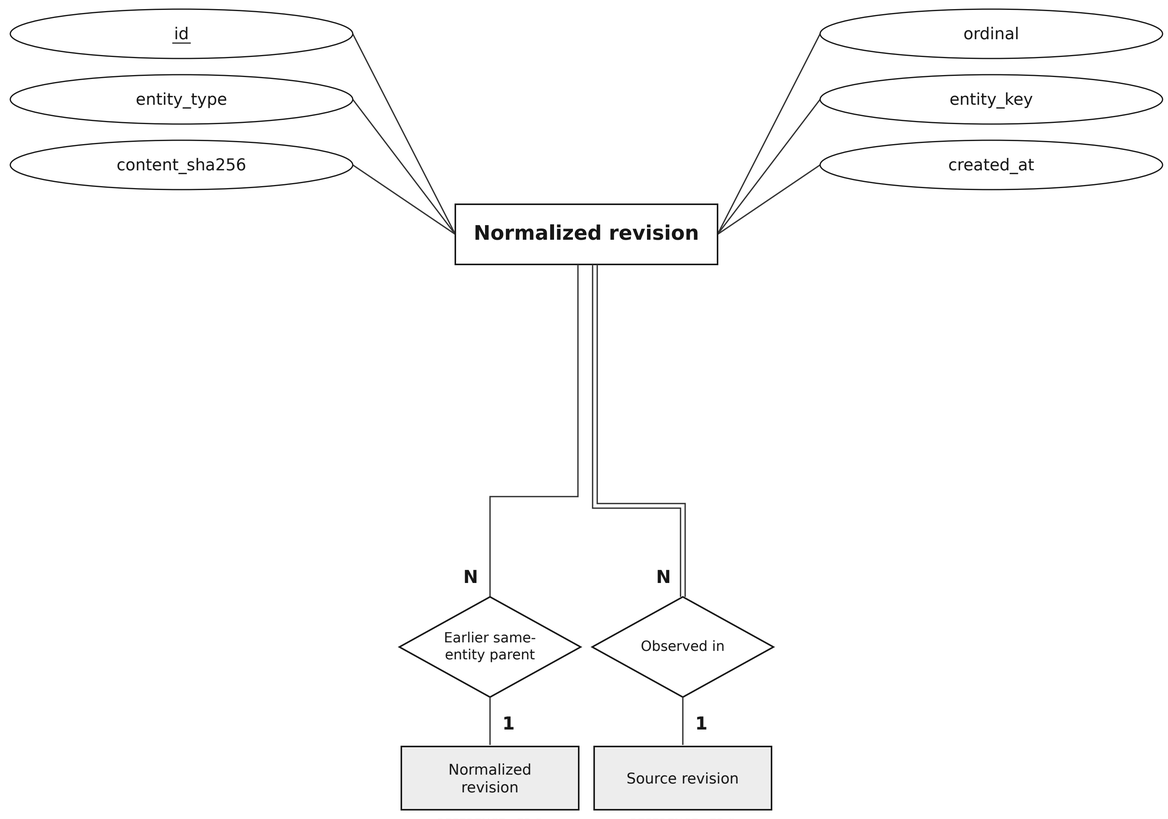
Entity and its stored scalar attributes



Independent entity: no declared relational parent.

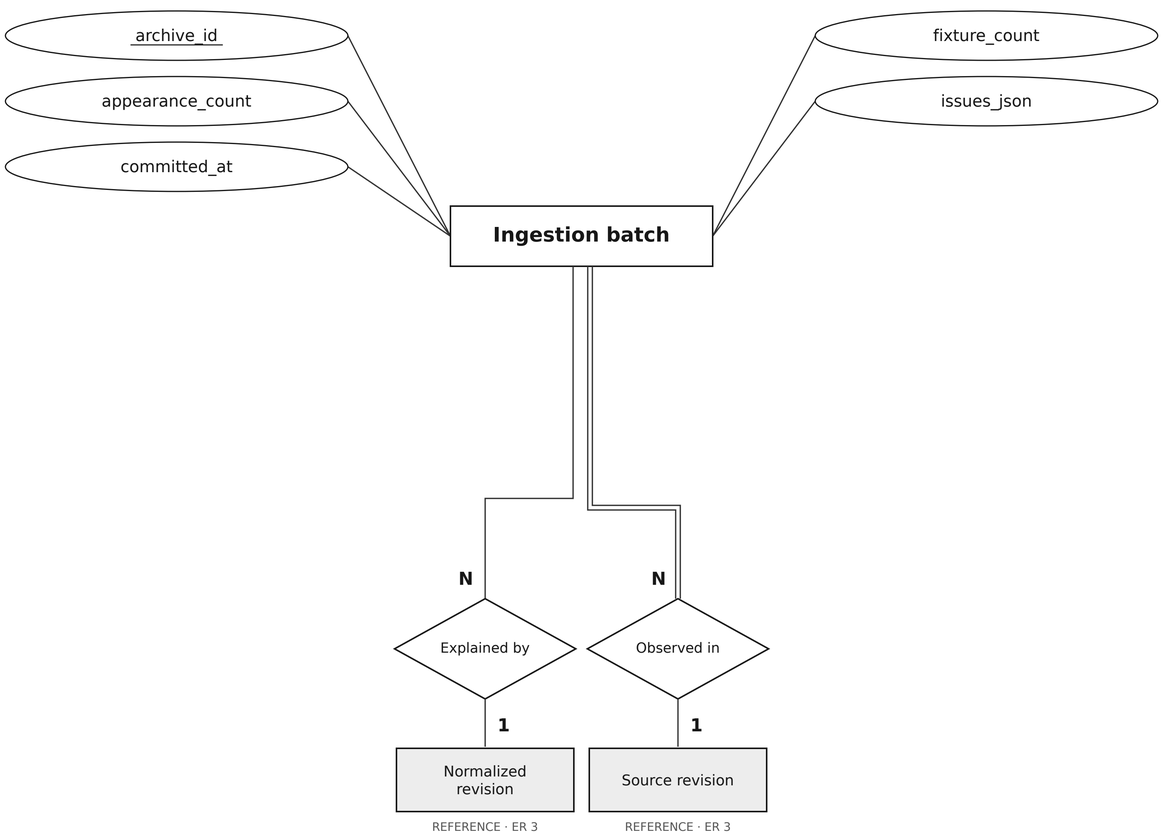
sports.normalized_revisions

Entity and its stored scalar attributes



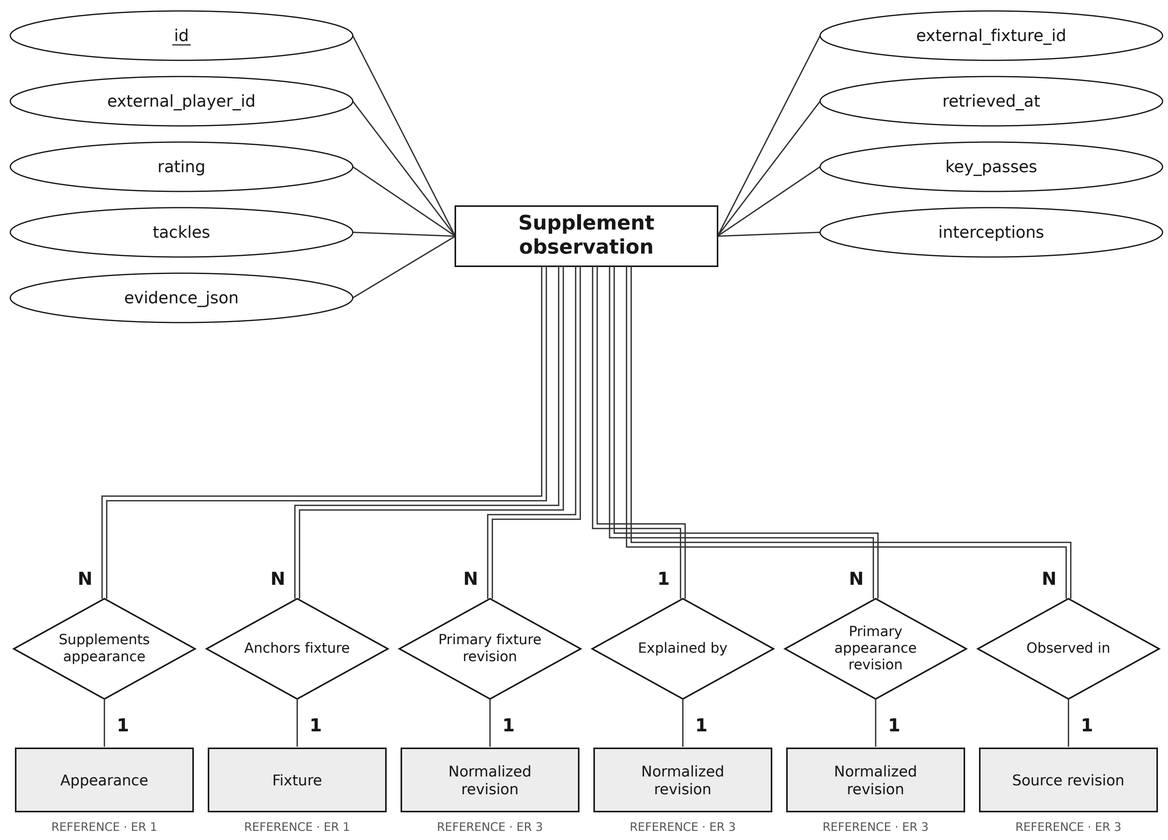
sports.ingestion_batches

Entity and its stored scalar attributes



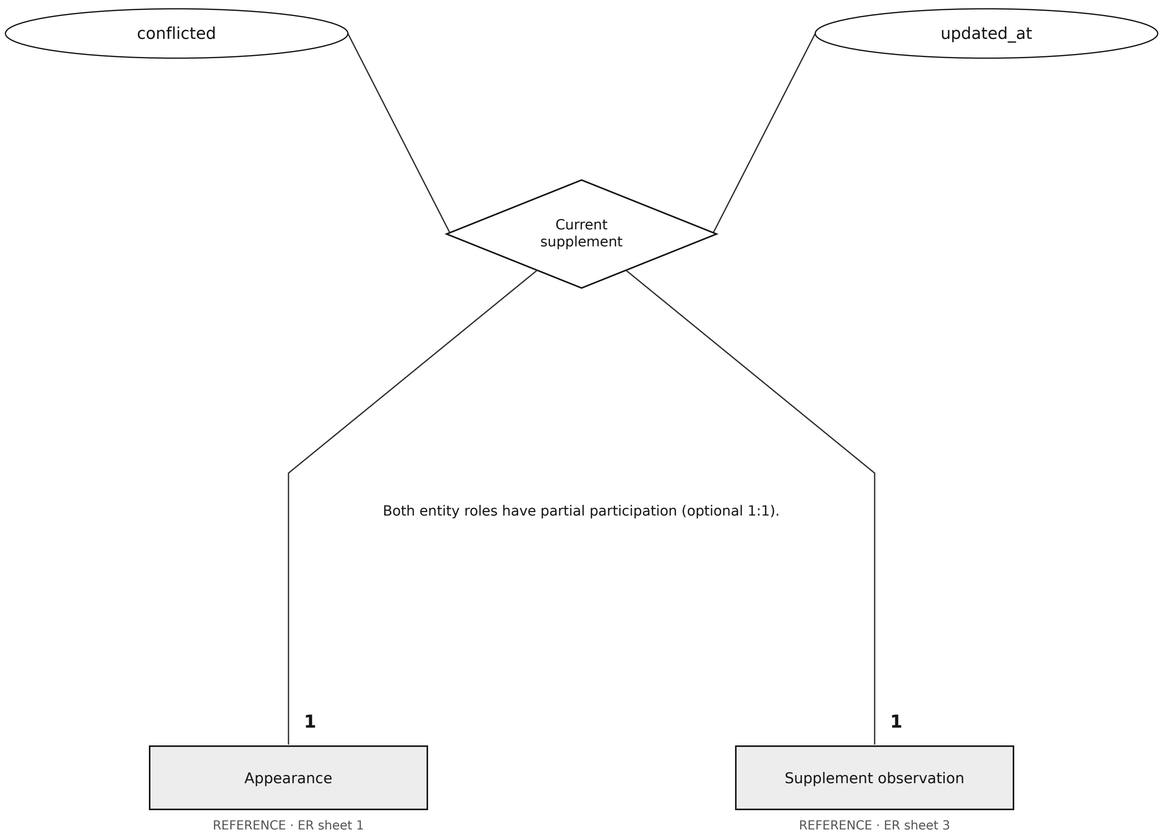
sports.appearance_enrichments

Entity and its stored scalar attributes



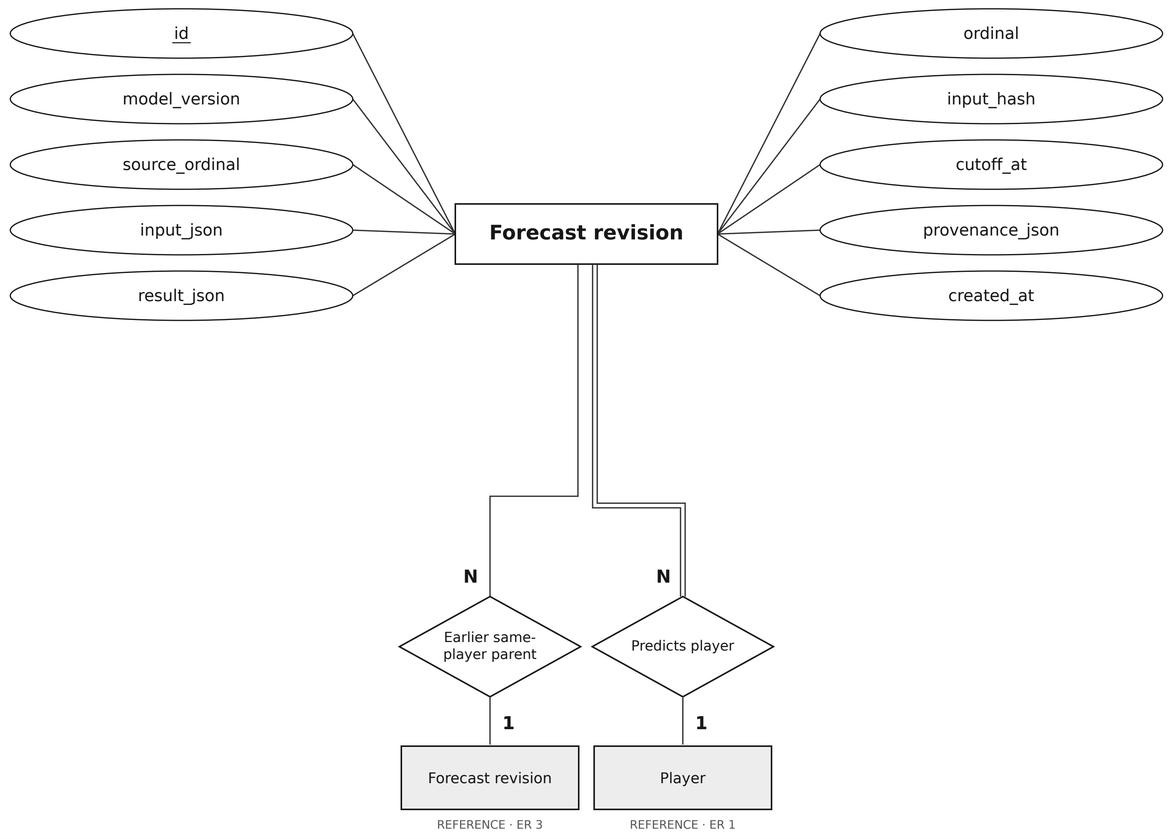
sports.appearance_enrichment_heads

Relationship attributes → head relation



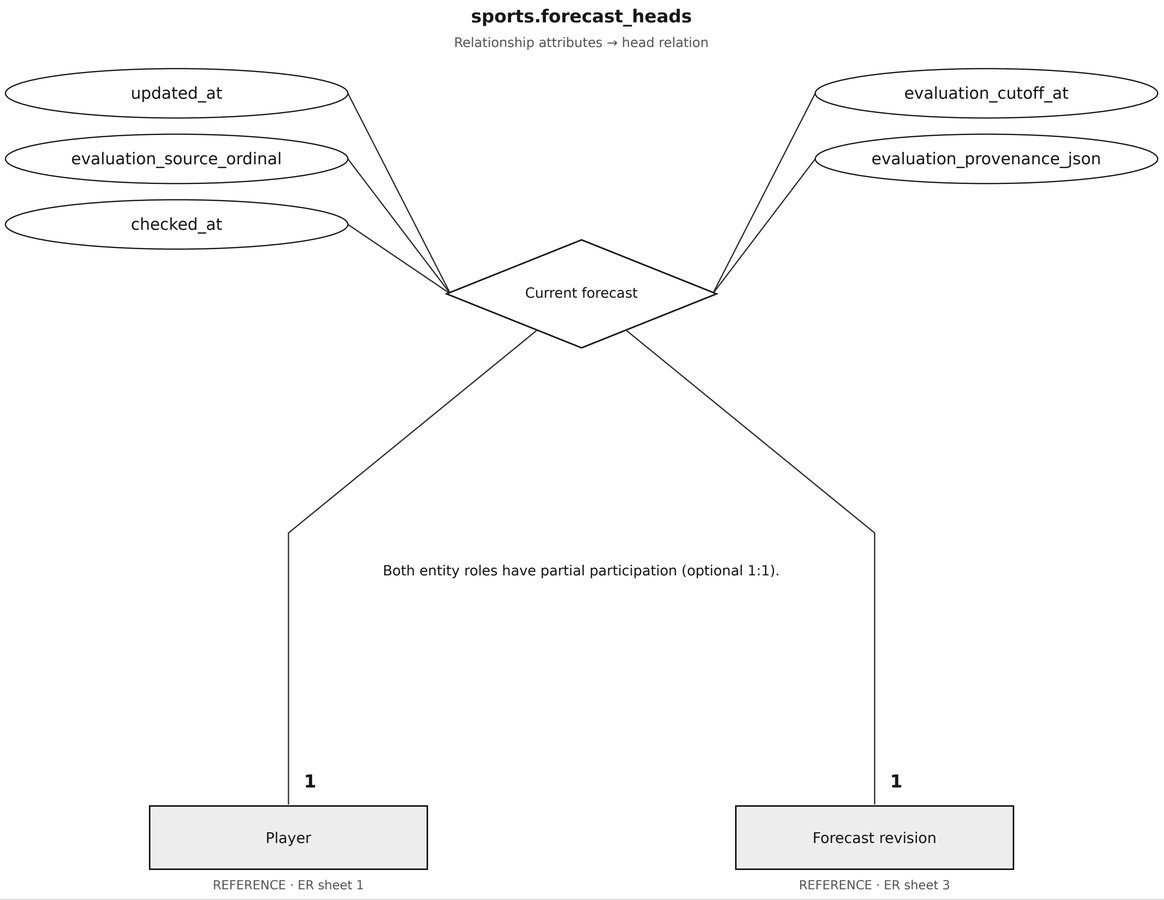
sports.forecast_revisions

Entity and its stored scalar attributes



sports.forecast_heads

Relationship attributes → head relation



Double rectangle + double diamond: weak identification. Dashed underline: partial key. Dashed oval: derived.

club_id identifies a standings row only within its owner snapshot; goal_difference = goals_for − goals_against.

These local views join on repeated entity names. FK column groups, alternate keys and conditional rules: relational-schema.md.

Current diamonds map to shared-owner-key head relations. No multivalued attribute or extra entity is invented.

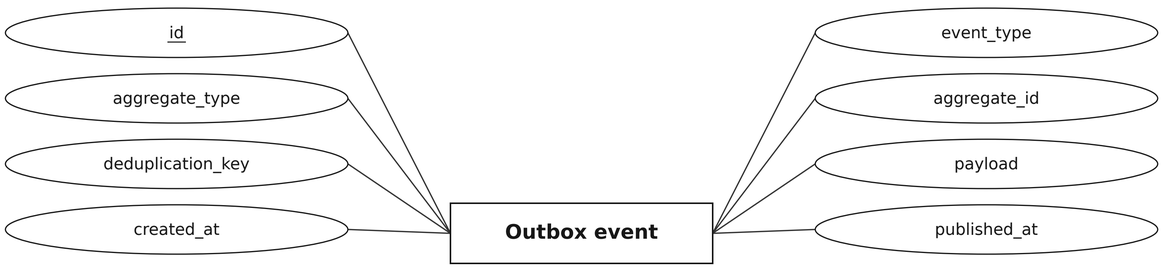
CHEN ER MODEL · 4/5 · Processing and infrastructure

Rectangles: entities · diamonds: relationships · ovals: attributes · solid underline: chosen key

1/N give maximum ratios; double lines give mandatory participation. Gray boxes repeat an existing entity.

sports.outbox

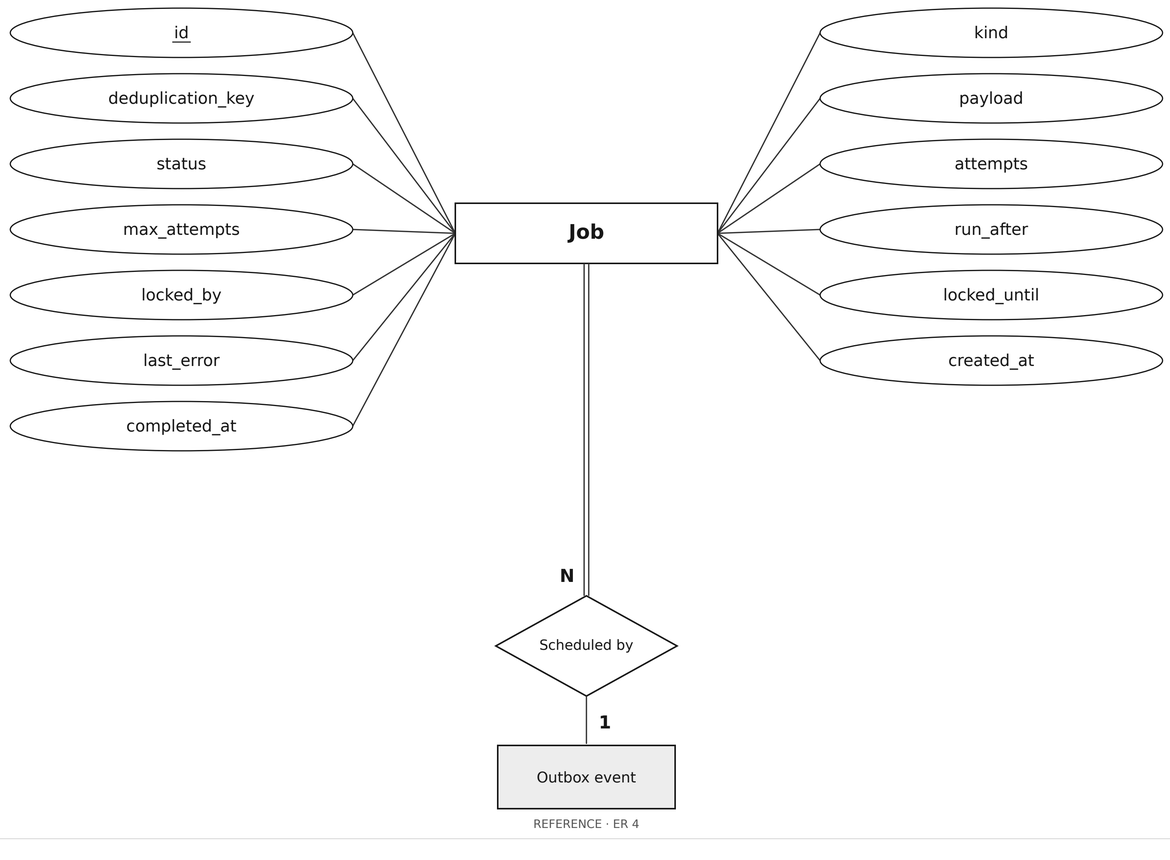
Entity and its stored scalar attributes



Independent entity: no declared relational parent.

sports.jobs

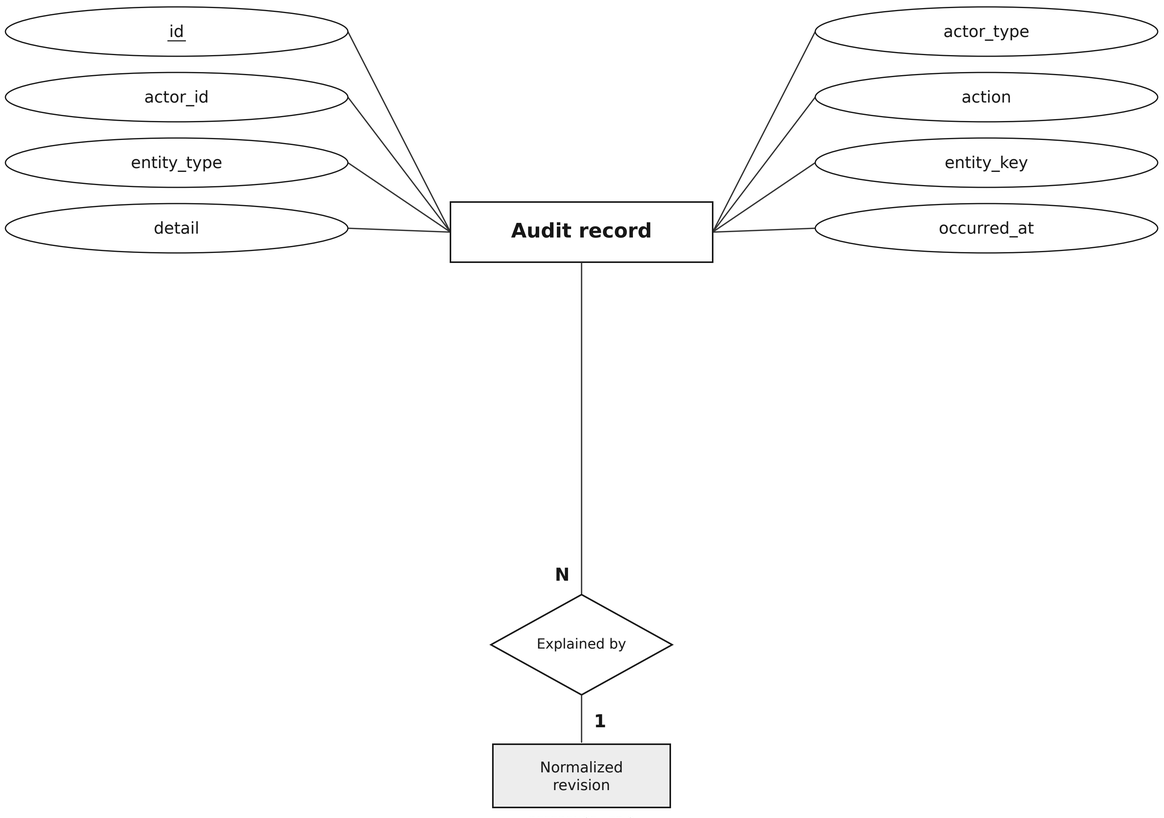
Entity and its stored scalar attributes



REFERENCE · ER 4

sports.audit_log

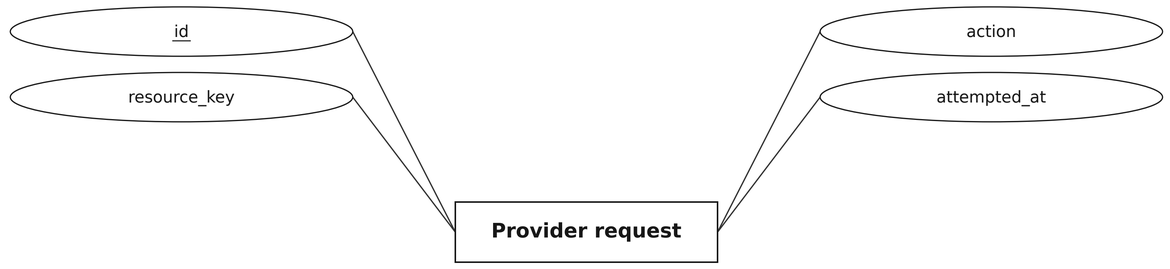
Entity and its stored scalar attributes



REFERENCE · ER 3

sports.provider_requests

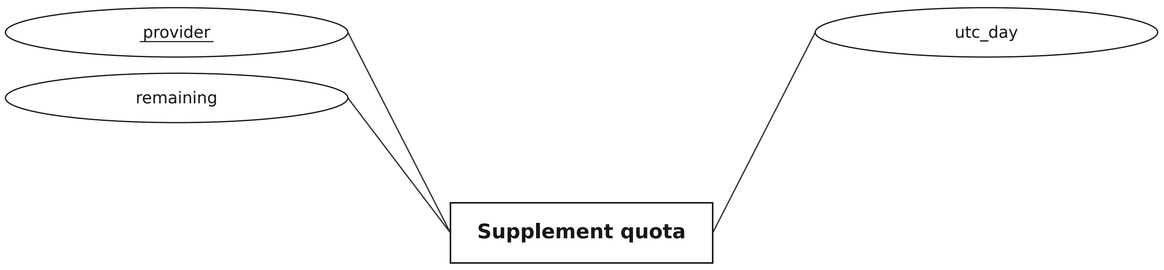
Entity and its stored scalar attributes



Independent entity: no declared relational parent.

sports.enrichment_request_quota

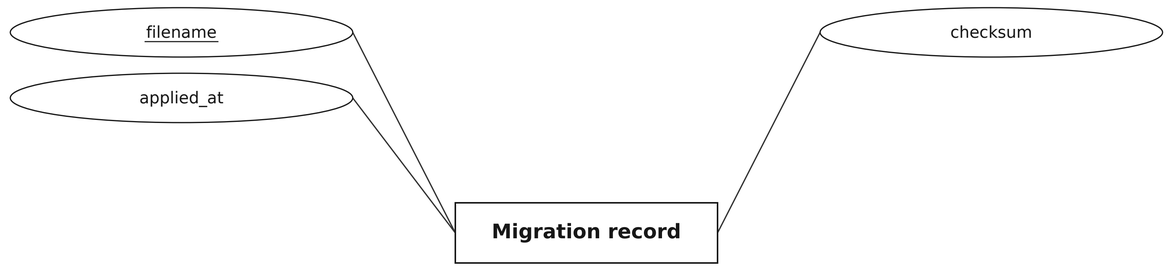
Entity and its stored scalar attributes



Independent entity: no declared relational parent.

infra.schema_migrations

Entity and its stored scalar attributes



Independent entity: no declared relational parent.

Double rectangle + double diamond: weak identification. Dashed underline: partial key. Dashed oval: derived.

club_id identifies a standings row only within its owner snapshot; goal_difference = goals_for − goals_against.

These local views join on repeated entity names. FK column groups, alternate keys and conditional rules: relational-schema.md.

Current diamonds map to shared-owner-key head relations. No multivalued attribute or extra entity is invented.

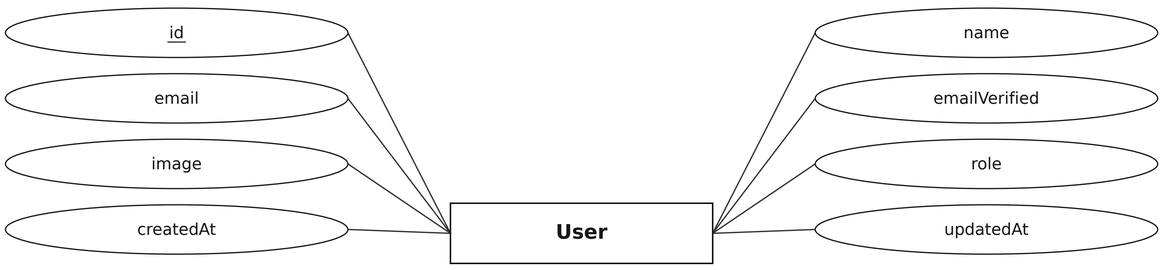
CHEN ER MODEL · 5/5 · Authentication storage

Rectangles: entities · diamonds: relationships · ovals: attributes · solid underline: chosen key

1/N give maximum ratios; double lines give mandatory participation. Gray boxes repeat an existing entity.

auth.user

Entity and its stored scalar attributes



Independent entity: no declared relational parent.

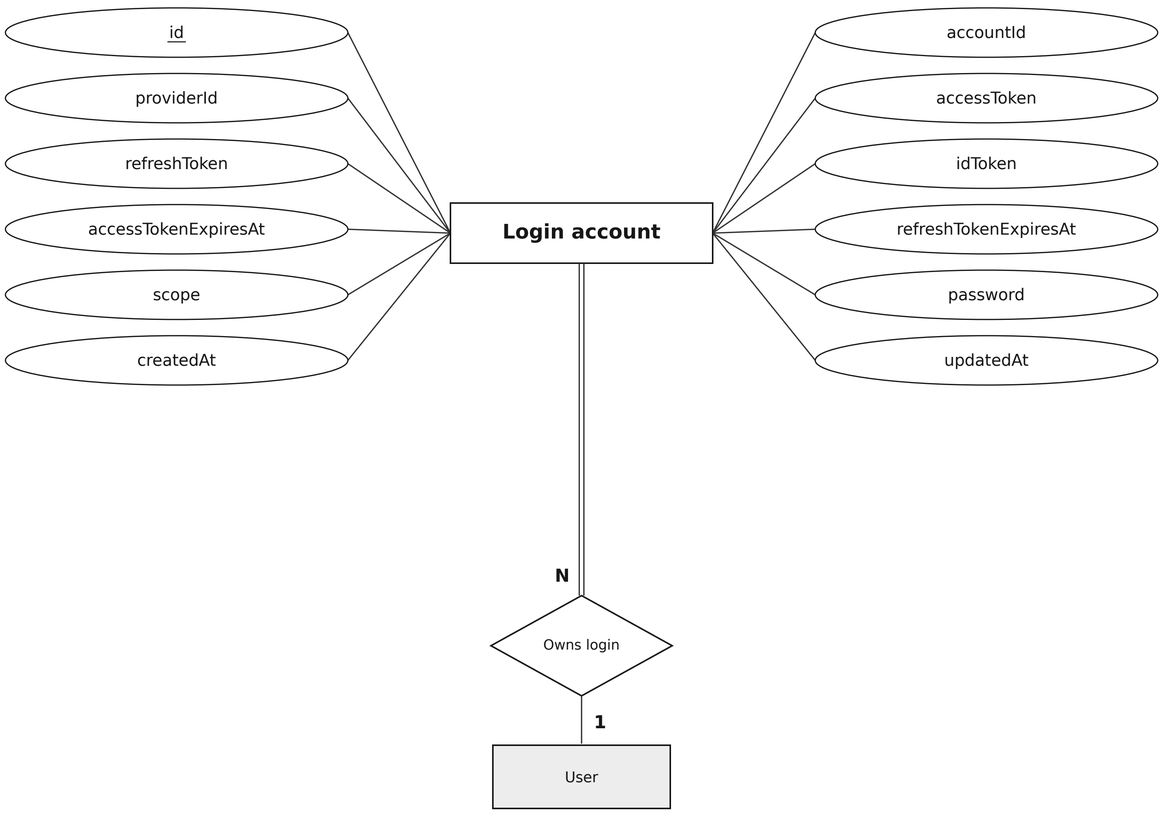
auth.session

Entity and its stored scalar attributes

REFERENCE · ER 5

auth.account

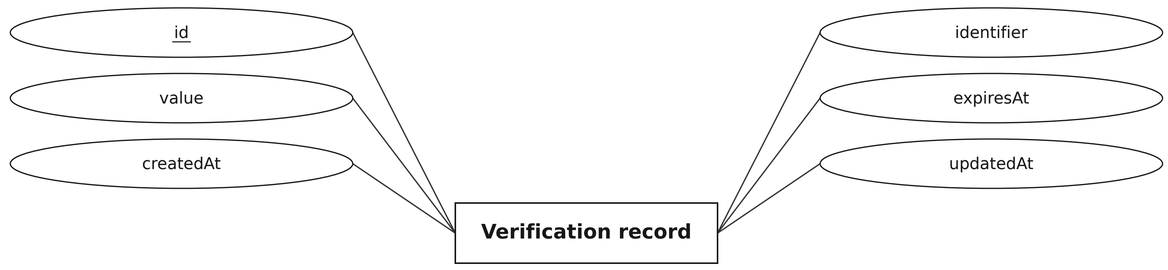
Entity and its stored scalar attributes



REFERENCE · ER 5

auth.verification

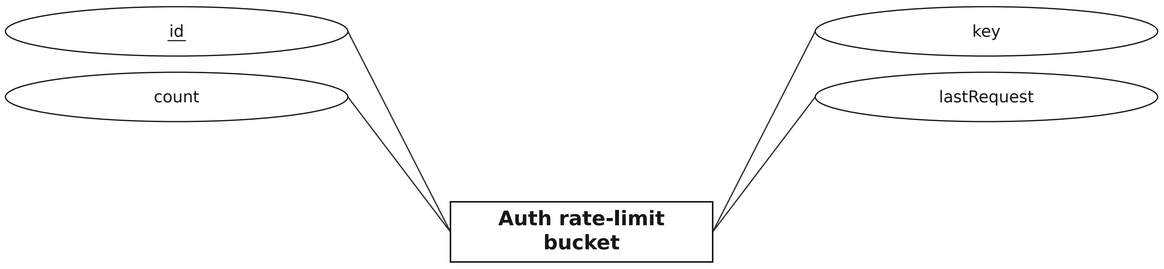
Entity and its stored scalar attributes



Independent entity: no declared relational parent.

auth.rateLimit

Entity and its stored scalar attributes



Independent entity: no declared relational parent.

Double rectangle + double diamond: weak identification. Dashed underline: partial key. Dashed oval: derived.

club_id identifies a standings row only within its owner snapshot; goal_difference = goals_for − goals_against.

These local views join on repeated entity names. FK column groups, alternate keys and conditional rules: relational-schema.md.

Current diamonds map to shared-owner-key head relations. No multivalued attribute or extra entity is invented.

RELATIONAL SCHEMA · 1/5 · Football and registration

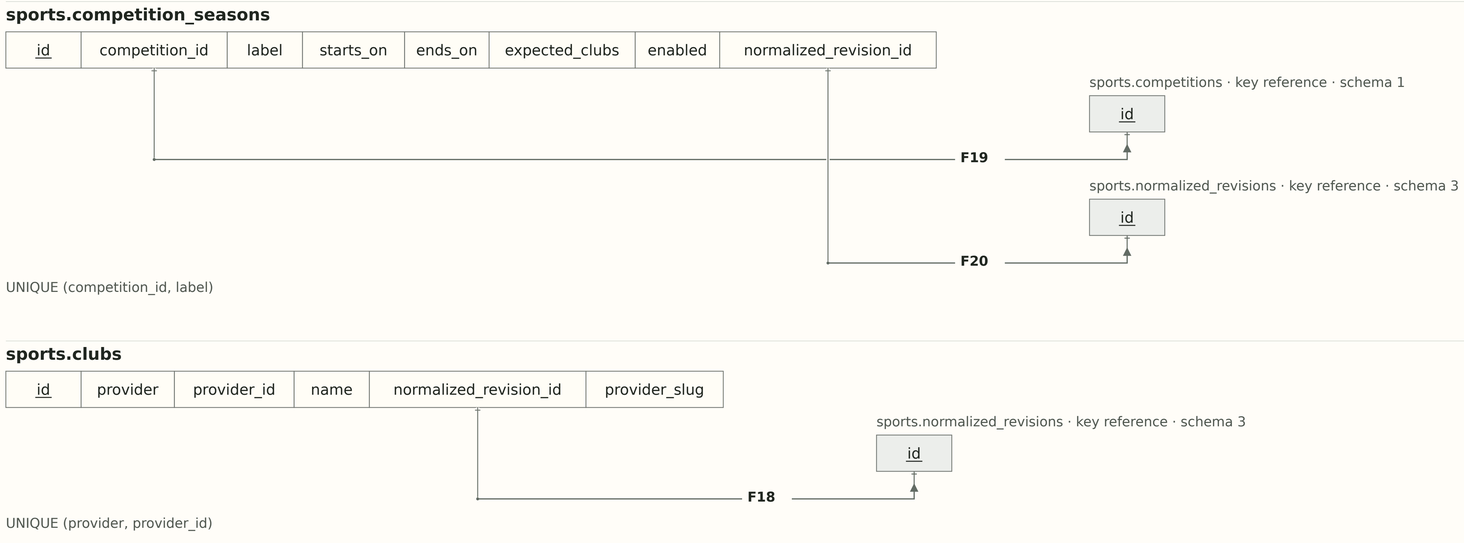
One named horizontal strip per relation. Underlined attributes form one combined primary key.
Each Fnn arrow represents one whole ordered FK tuple; joined ticks show every source and target component.
Gray strips repeat only referenced key attributes. Types, nullability and full constraints: relational-schema.md.

sports.competitions



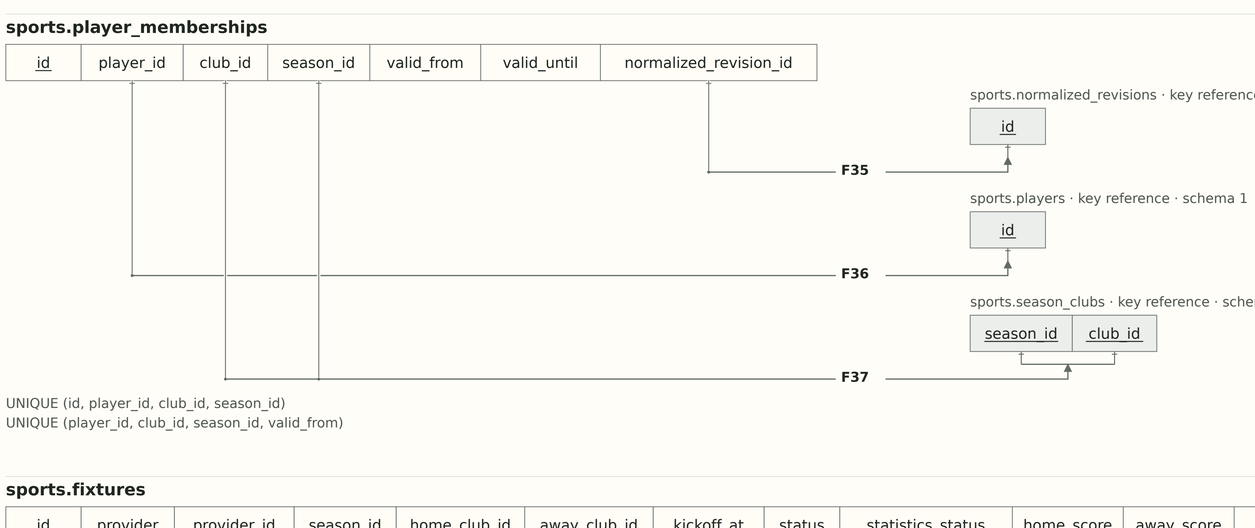
UNIQUE (provider, provider_id)

sports.competition_seasons



UNIQUE (competition_id, label)

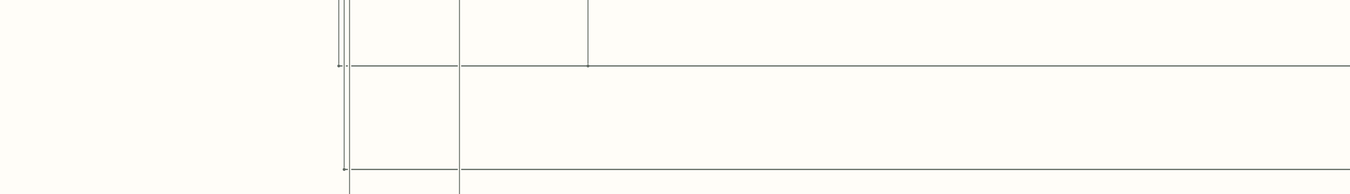
sports.clubs



UNIQUE (provider, provider_id)

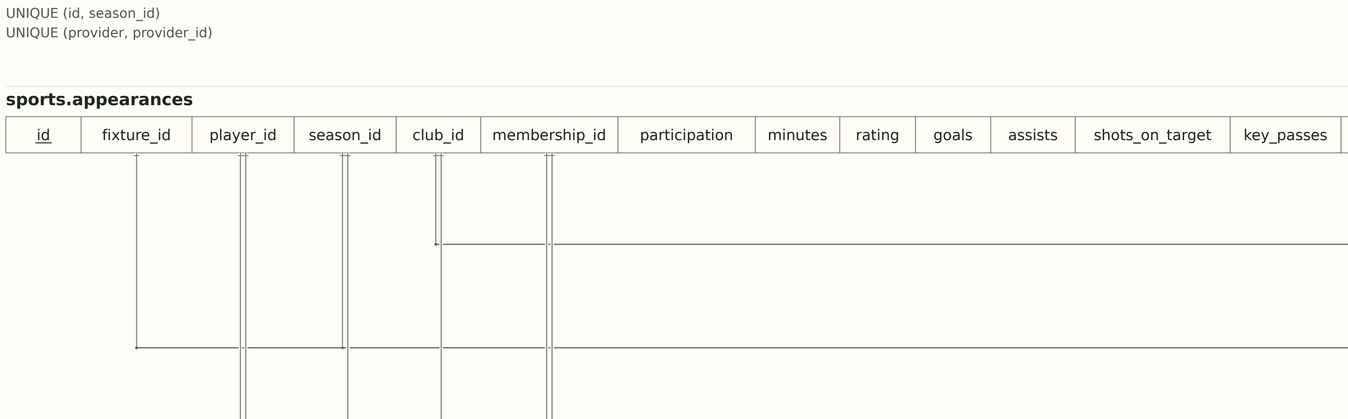
sports.season_clubs

sports.players



UNIQUE (provider, provider_id)

sports.player_memberships



UNIQUE (id, player_id, club_id, season_id)
UNIQUE (player_id, club_id, season_id, valid_from)

sports.fixtures

UNIQUE (id, season_id)

UNIQUE (provider, provider_id)

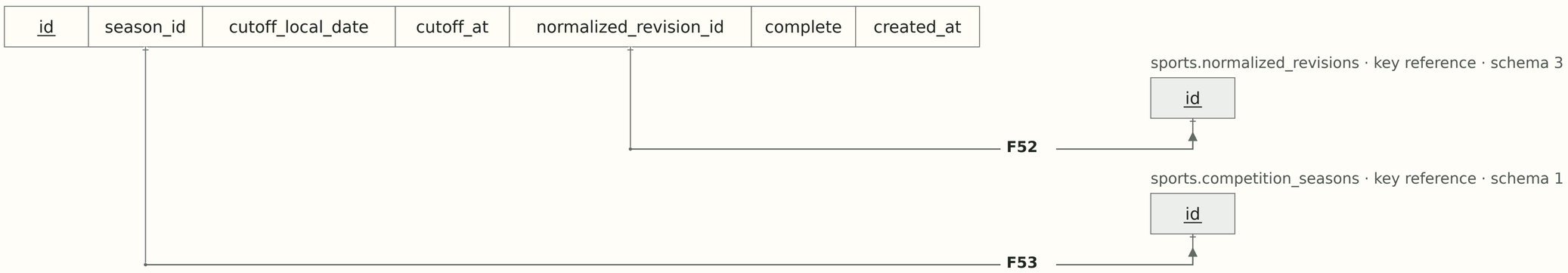
sports.appearances

UNIQUE (fixture_id, player_id)

RELATIONAL SCHEMA · 2/5 · Standings and roster observations

One named horizontal strip per relation. Underlined attributes form one combined primary key.
Each Fnn arrow represents one whole ordered FK tuple; joined ticks show every source and target component.
Gray strips repeat only referenced key attributes. Types, nullability and full constraints: relational-schema.md.

sports.standings_snapshots

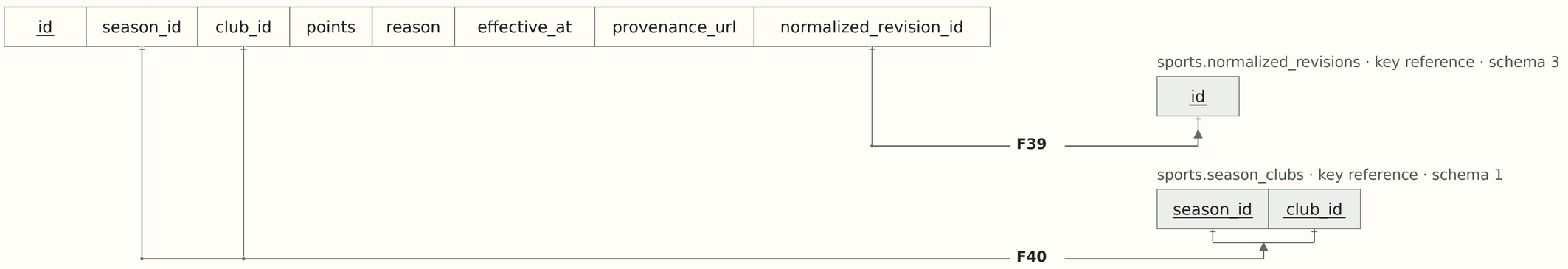


UNIQUE (id, season_id)
UNIQUE (season_id, cutoff_local_date, normalized_revision_id)

sports.standings_rows

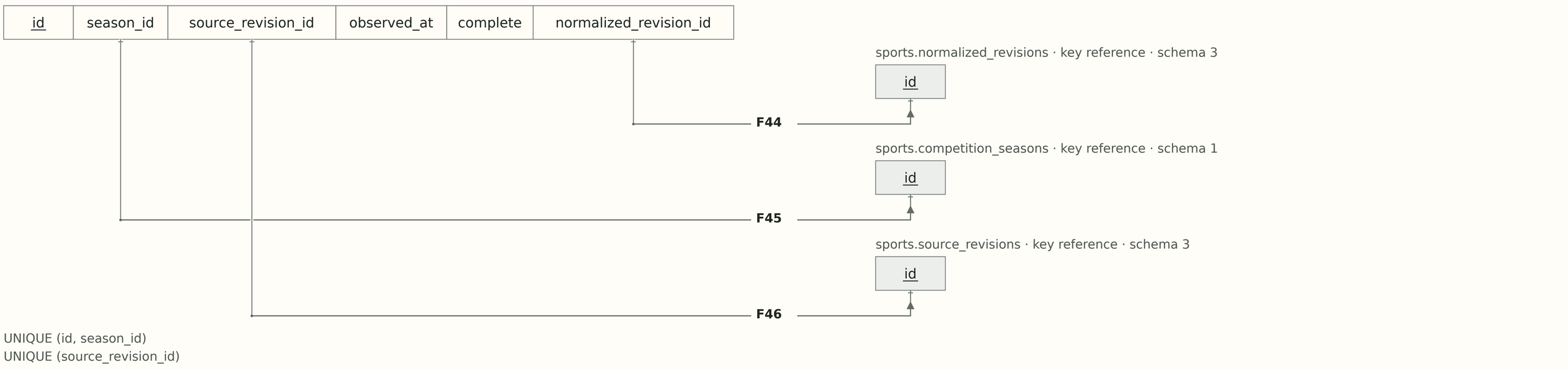


sports.points_adjustments



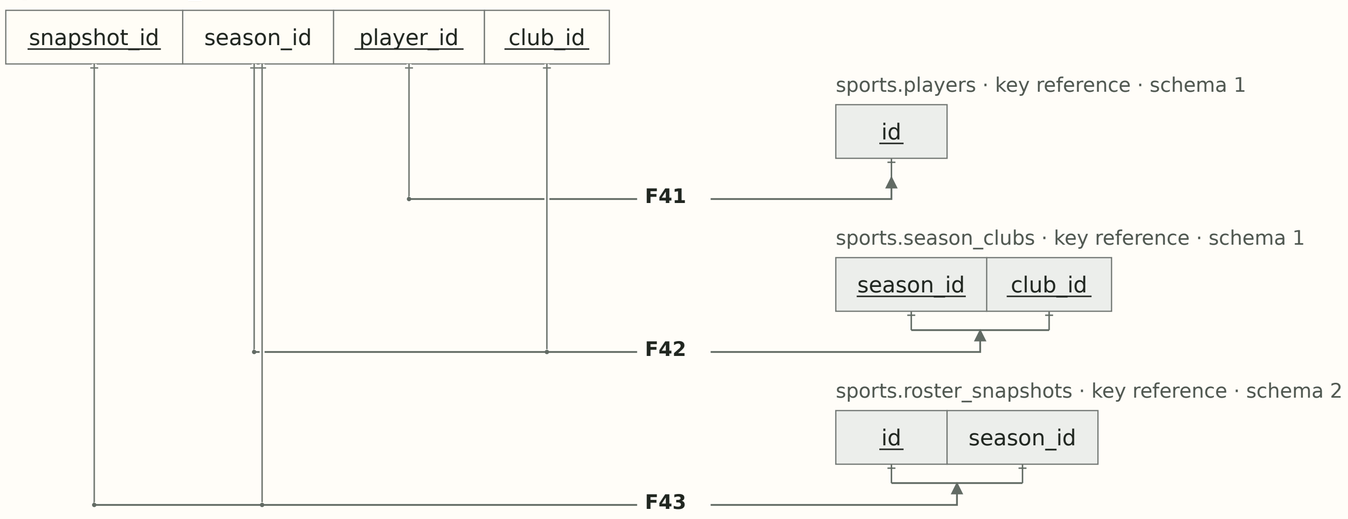
UNIQUE (season_id, club_id, effective_at, provenance_url)

sports.roster_snapshots



UNIQUE (id, season_id)
UNIQUE (source_revision_id)

sports.roster_entries



RELATIONAL SCHEMA · 3/5 · Provenance, supplements and forecasts

One named horizontal strip per relation. Underlined attributes form one combined primary key.
Each Fnn arrow represents one whole ordered FK tuple; joined ticks show every source and target component.
Gray strips repeat only referenced key attributes. Types, nullability and full constraints: relational-schema.md.

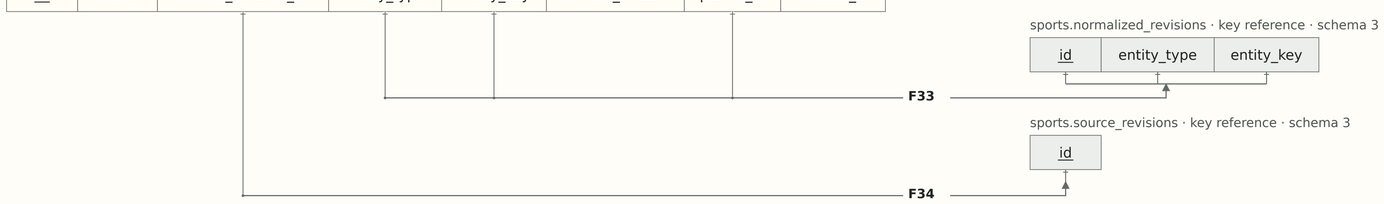
sports.source_revisions

id	provider	action	resource_key	archive_ref	content_sha256	retrieved_at

UNIQUE (archive_ref)
UNIQUE (provider, action, resource_key, content_sha256, retrieved_at)

sports.normalized_revisions

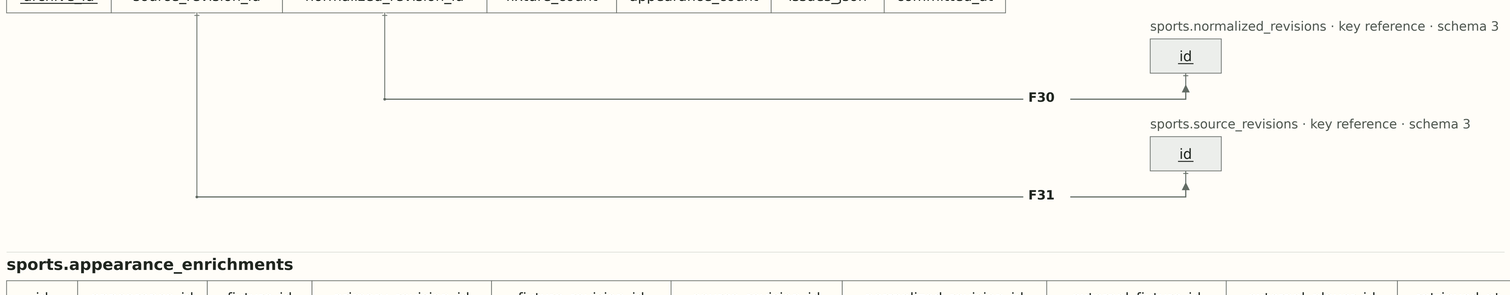
id	ordinal	source_revision_id	entity_type	entity_key	content_sha256	parent_id	created_at



UNIQUE (id, entity_type, entity_key)
UNIQUE (ordinal)
UNIQUE (source_revision_id, entity_type, entity_key, content_sha256)

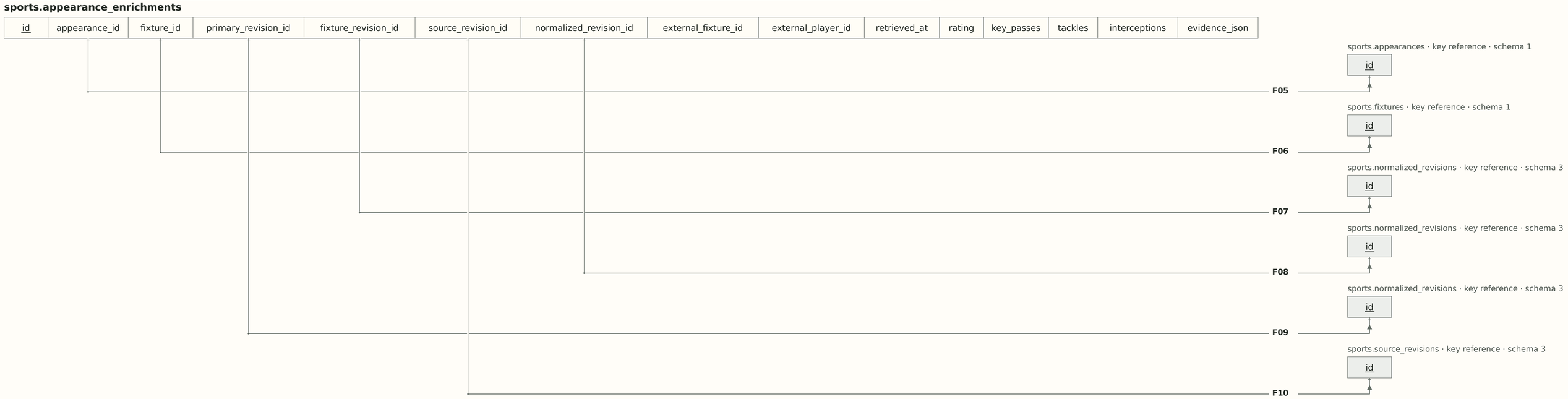
sports.ingestion_batches

archive_id	source_revision_id	normalized_revision_id	fixture_count	appearance_count	issues_json	committed_at



sports.appearance_enrichments

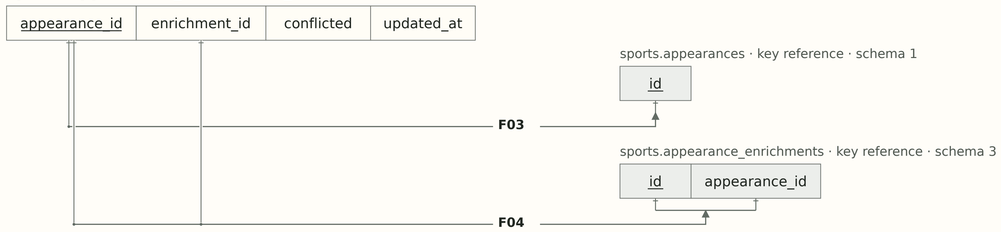
id	appearance_id	fixture_id	primary_revision_id	fixture_revision_id	source_revision_id	normalized_revision_id	external_fixture_id	external_player_id	retrieved_at	rating	key_passes	tackles	interceptions	evidence_json



UNIQUE (id, appearance_id)
UNIQUE (normalized_revision_id)
UNIQUE (source_revision_id, appearance_id)

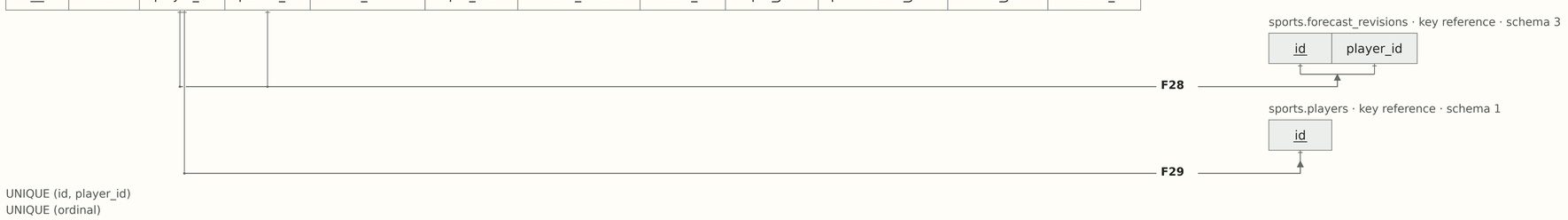
sports.appearance_enrichment_heads

appearance_id	enrichment_id	conflicted	updated_at



sports.forecast_revisions

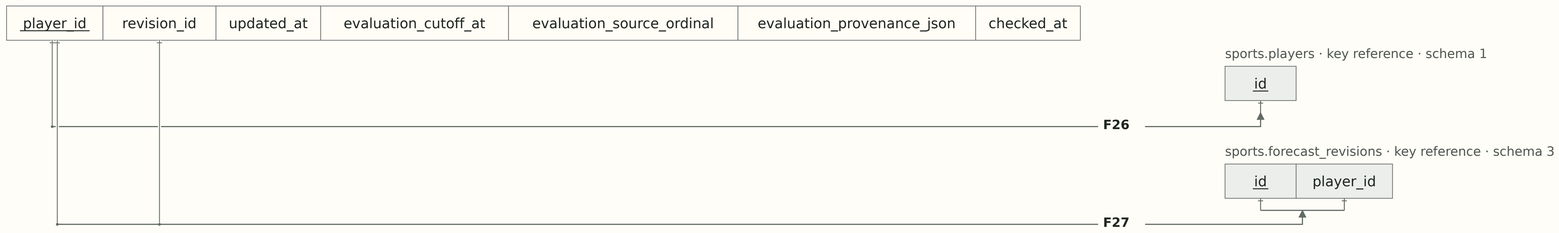
id	ordinal	player_id	parent_id	model_version	input_hash	source_ordinal	cutoff_at	input_json	provenance_json	result_json	created_at



UNIQUE (id, player_id)
UNIQUE (ordinal)

sports.forecast_heads

player_id	revision_id	updated_at	evaluation_cutoff_at	evaluation_source_ordinal	evaluation_provenance_json	checked_at



RELATIONAL SCHEMA · 4/5 · Processing and infrastructure

One named horizontal strip per relation. Underlined attributes form one combined primary key.
Each Fnn arrow represents one whole ordered FK tuple; joined ticks show every source and target component.
Gray strips repeat only referenced key attributes. Types, nullability and full constraints: relational-schema.md.

sports.outbox

id	event_type	aggregate_type	aggregate_id	deduplication_key	payload	created_at	published_at

UNIQUE (deduplication_key)

sports.jobs

id	outbox_id	kind	deduplication_key	payload	status	attempts	max_attempts	run_after	locked_by	locked_until	last_error	created_at	completed_at

sports.outbox · key reference · schema 4

id

F32

UNIQUE (deduplication_key)

sports.audit_log

id	normalized_revision_id	actor_type	actor_id	action	entity_type	entity_key	detail	occurred_at

sports.normalized_revisions · key reference · schema 3

id

F17

sports.provider_requests

id	action	resource_key	attempted_at

No declared foreign keys or alternate UNIQUE constraints.

sports.enrichment_request_quota

provider	utc_day	remaining

No declared foreign keys or alternate UNIQUE constraints.

infra.schema_migrations

filename	checksum	applied_at

No declared foreign keys or alternate UNIQUE constraints.

RELATIONAL SCHEMA · 5/5 · Authentication storage

One named horizontal strip per relation. Underlined attributes form one combined primary key.
Each Fnn arrow represents one whole ordered FK tuple; joined ticks show every source and target component.
Gray strips repeat only referenced key attributes. Types, nullability and full constraints: relational-schema.md.

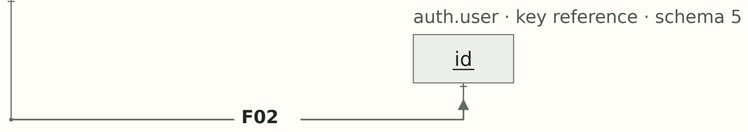
auth.user

id	name	email	emailVerified	image	role	createdAt	updatedAt

UNIQUE INDEX (lower(email))

auth.session

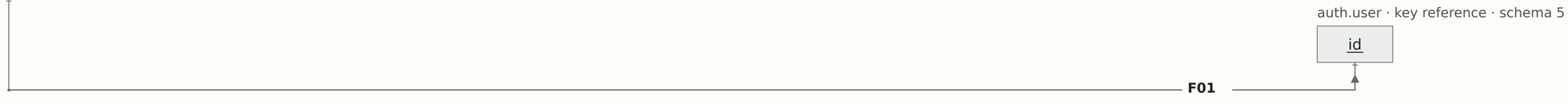
id	expiresAt	token	createdAt	updatedAt	ipAddress	userAgent	userId	persistent



UNIQUE (token)

auth.account

id	accountId	providerId	userId	accessToken	refreshToken	idToken	accessTokenExpiresAt	refreshTokenExpiresAt	scope	password	createdAt	updatedAt



UNIQUE (providerId, accountId)

auth.verification

id	identifier	value	expiresAt	createdAt	updatedAt

No declared foreign keys or alternate UNIQUE constraints.

auth.rateLimit

id	key	count	lastRequest

UNIQUE (key)